



DEPARTMENT OF MATERIAL SCIENCE AND ENGINEERING
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THERMODYNAMIC ANALYSIS OF A HEAT SINK

MT1070 - THERMODYNAMICS AND PHASE EQUILIBRIA

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THERMODYNAMIC ANALYSIS OF A HEAT SINK

1. INTRODUCTION

Heat sinks are vital components in various industries, playing a crucial role in dissipating excess heat generated by electronic devices and machinery. They utilize multiple mechanisms, including conduction, convection, and radiation, to efficiently transfer thermal energy away from the heat source and maintain optimal operating temperatures. This report aims to provide an in-depth understanding of heat sink technology, focusing on the mechanisms employed, modifications implemented, and their impact on overall performance.

The conduction mechanism forms the foundation of heat sink operation, involving the transfer of heat through direct contact between two solids. It relies on high thermal conductivity materials, such as copper or aluminum, to efficiently conduct heat away from the heat-generating component. The heat is then distributed across the heat sink's surface area, allowing for effective dissipation.

Convection, on the other hand, utilizes the movement of fluid (liquid or gas) to carry away heat. This mechanism significantly enhances heat transfer efficiency, as it takes advantage of the fluid's ability to absorb and transport heat. The heat sink's design incorporates fins or extended surfaces to increase the available surface area for convective heat transfer, facilitating faster cooling.

In addition to conduction and convection, heat sinks also employ radiation to dissipate heat. Radiation is the transfer of heat energy through electromagnetic waves, primarily in the form of infrared radiation. Heat sinks often feature surfaces with high emissivity to maximize the radiation effect, allowing the emitted heat to escape into the surrounding environment.

To optimize heat sink performance, various modifications can be implemented. These include the design and arrangement of fins, selection of appropriate materials, incorporation of heat pipes, and integration of advanced cooling techniques such as liquid cooling or vapor chambers. Each modification aims to enhance the heat sink's ability to dissipate heat efficiently, thus preventing thermal damage and improving overall system reliability.

Throughout this report, we will explore these mechanisms and modifications in detail, analyzing their impact on heat sink performance. By understanding the underlying principles and advancements in heat sink technology, we can pave the way for further improvements in thermal management and contribute to the development of more efficient cooling solutions.

2. HEAT TRANSFER MECHANISMS

Understanding and discussing the heat transfer mechanisms when conducting a thermal analysis of a heat sink is of utmost importance. These mechanisms—conduction, convection, and radiation—play a crucial role in determining the heat sink's effectiveness in dissipating heat from the component it is cooling. By considering conduction, engineers can optimize the thermal interface between the heat sink and the component, reducing thermal resistance and improving heat transfer. Discussing convection allows for the evaluation of airflow patterns, fin design, and fan placement, optimizing the heat sink's ability to carry heat away from its surface. Additionally, considering radiation enables engineers to assess the impact of electromagnetic waves on heat transfer, especially at high temperatures. By analyzing and discussing these heat transfer mechanisms, engineers can design and optimize heat sinks for efficient thermal management, preventing overheating and enhancing component performance and reliability.

2.1. CONDUCTION PROCESS IN A HEAT SINK

Conduction is the process of heat transfer through direct contact between two objects. In a heat sink, the heat from the hot source is transferred to the heat sink fins through conduction. Then, the fins transfer the heat to the surrounding air or fluid through convection and radiation.

2.1.1. THE TEMPERATURE FIELD

A temperature field is defined as,

$$T = f(x, y, z, t) \text{ -----(2.1.1.1)}$$

If temperature field is steady,

$$T = f(x, y, z) \text{ -----(2.1.1.2)}$$

If temperature field is a two dimensional,

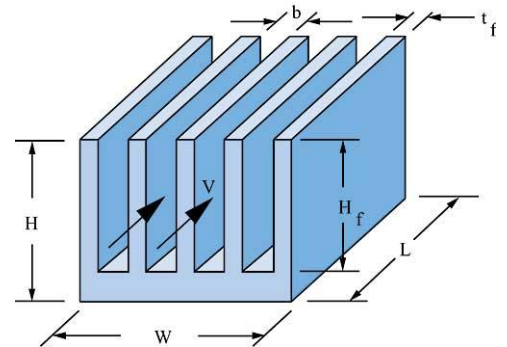
$$T = f(x, y, t), \frac{\partial T}{\partial z} = 0 \text{ -----(2.1.1.3)}$$

If temperature field is a one dimensional,

$$T = f(x, t), \frac{\partial T}{\partial z} = 0, \frac{\partial T}{\partial y} = 0 \text{ -----(2.1.1.4)}$$

If temperature field is a one dimensional and steady,

$$T = f(x), \frac{\partial T}{\partial z} = 0, \frac{\partial T}{\partial y} = 0, \frac{\partial T}{\partial t} = 0 \text{ -----(2.1.1.5)}$$



The locus of the curve joining the same temperature at all points of a body is called an isotherm. Figure 2.1.1.1 shows isotherms differing by temperature ΔT . The isotherms or isothermal surfaces never interact. They either end at the surface of the body or are located completely within it. The temperature of a body can vary only in directions crossing isotherms. **The greatest temperature variation** is in a direction normal to the isothermal surface, and it is defined as the temperature gradient.

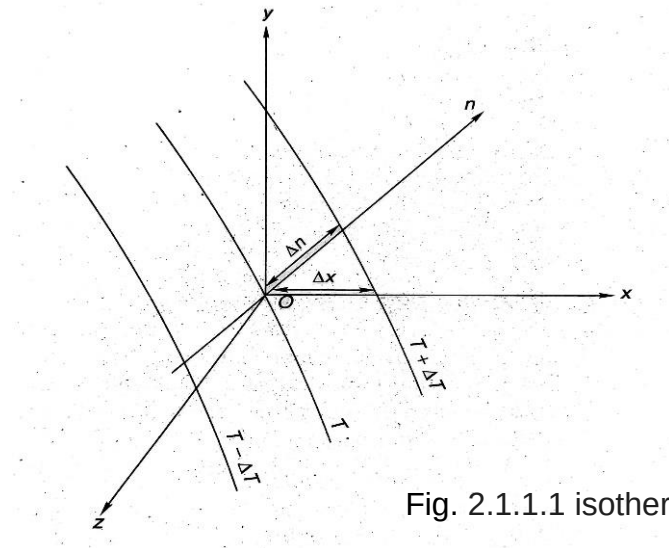


Fig. 2.1.1.1 isothermal surfaces

$$\nabla T = n \frac{\partial T}{\partial n} \text{-----(2.1.1.6)}$$

- ∇T - the temperature gradient
- n - the unit vector normal to the isotherm
- $\frac{\partial T}{\partial n}$ - the temperature derivative along the normal n

* $\frac{\partial T}{\partial n} > 0$ in the the direction of increasing temperature

The vector eqn. (2.1.1.6) can be written in scalar for the coordinate axes ox , oy , oz ,

$$\frac{\partial T}{\partial x} = l \frac{\partial T}{\partial n}, \frac{\partial T}{\partial y} = m \frac{\partial T}{\partial n}, \frac{\partial T}{\partial z} = n \frac{\partial T}{\partial n} \text{-----(2.1.1.7)}$$

where l, m, n are the direction cosines.

2.1.2. CONDUCTION

Fourier's law in differential form: $q = -k\nabla T$ -----(2.1.2.1)

Where,

- q is the local heat flux density in Wm^2
- k is the coefficient of thermal conductivity of the material in $\text{Wm}^{-1}\text{K}^{-1}$
- ∇T is the temperature gradient in Km^{-1}

The heat flux is the quantity of heat transferred per unit time per unit area of an isothermal surface.

The heat flux, q , is normal to the isothermal surface, and is positive in the direction of decreasing temperature because according to the second law of thermodynamics, heat always flows from a hotter point to a colder one. Hence, the vectors ∇T and q are both normal to isotherms but run in opposite directions as shown in Fig. 2.1.2.1.

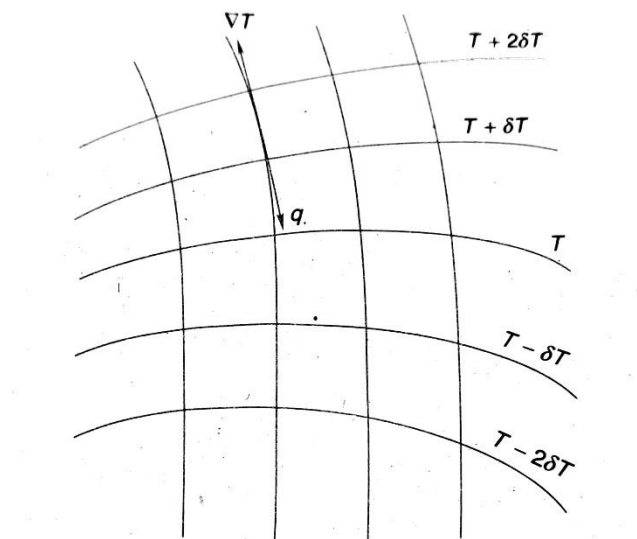


Fig 2.1.2.1 Lines of Heat flow and isotherms

According to Fourier's law, the rate of heat conduction is proportional to the area measured normal to the direction of heat flow, and to the temperature gradient in that direction.

$$Q = -kA \frac{\partial T}{\partial n} \text{ -----(2.1.2.2)}$$

* Q - the rate of heat conduction

The greatest rate of heat flow will be along the lines normal to the isothermal surfaces. Then,

$$Q_x = -k_x A \frac{\partial T}{\partial x} \quad \text{or} \quad q_x = \frac{Q_x}{A} = -k_x \frac{\partial T}{\partial x}$$

$$Q_y = -k_y A \frac{\partial T}{\partial y} \quad \text{or} \quad q_y = \frac{Q_y}{A} = -k_y \frac{\partial T}{\partial y} \quad \text{-----}(2.1.2.2)$$

$$Q_z = -k_z A \frac{\partial T}{\partial z} \quad \text{or} \quad q_z = \frac{Q_z}{A} = -k_z \frac{\partial T}{\partial z}$$

A material having $k_x = k_y = k_z = k$ is called an isotropic material. For isotropic material,

$$Q_x = -kA \frac{\partial T}{\partial x} \quad \text{-----}(2.1.2.3)$$

For any material the heat flow vector q can be written as,

$$\mathbf{q} = i q_x + j q_y + k q_z \quad \text{-----}(2.1.2.4)$$

The general nomenclature for heat flow in x-direction is shown in fig.2.1.2.2

For a plane wall of thickness L with temperatures T_0 and T_L on its two sides, integration of eqn. 2.1.2.3

$$Q_x \int_0^L dx = -kA \int_{T_0}^{T_L} dT \quad \text{or} \quad Q_x = \frac{kA}{L} (T_0 - T_L) \quad \text{-----}(2.1.2.5)$$

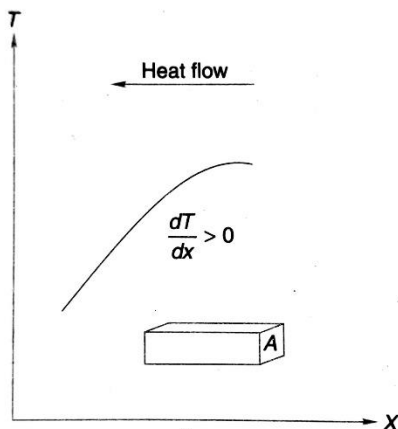


Fig 2.1.2.3. one dimensional conduction

The unit of the thermal conductivity, k - $\text{W/m}^\circ\text{C}$ or W/mK

the heat flux, q - W/m^2

the temperature gradient - $^\circ\text{C/m}$

2.1.3. GENERAL DIFFERENTIAL EQUATION OF HEAT CONDUCTION

Consider a small volume element in Cartesian coordinates having sides dx , dy and dz as shown in Fig. 2.1.3.1. The energy balance for this little element is obtained from the first law of thermodynamics as:

$$\begin{array}{ccccccc}
 \begin{array}{c} \text{Net heat} \\ \text{conducted into} \\ \text{element} \\ dx \, dy \, dz \text{ per} \\ \text{unit time} \\ \text{(I)} \end{array} & + & \begin{array}{c} \text{Internal heat} \\ \text{generated per} \\ \text{unit time} \\ \text{(II)} \end{array} & = & \begin{array}{c} \text{Increase} \\ \text{internal energy} \\ \text{per unit time} \\ \text{(III)} \end{array} & + & \begin{array}{c} \text{Work done by} \\ \text{element per} \\ \text{unit time} \\ \text{(IV)} \end{array} & \text{-----} & \text{(2.1.3.1)}
 \end{array}$$

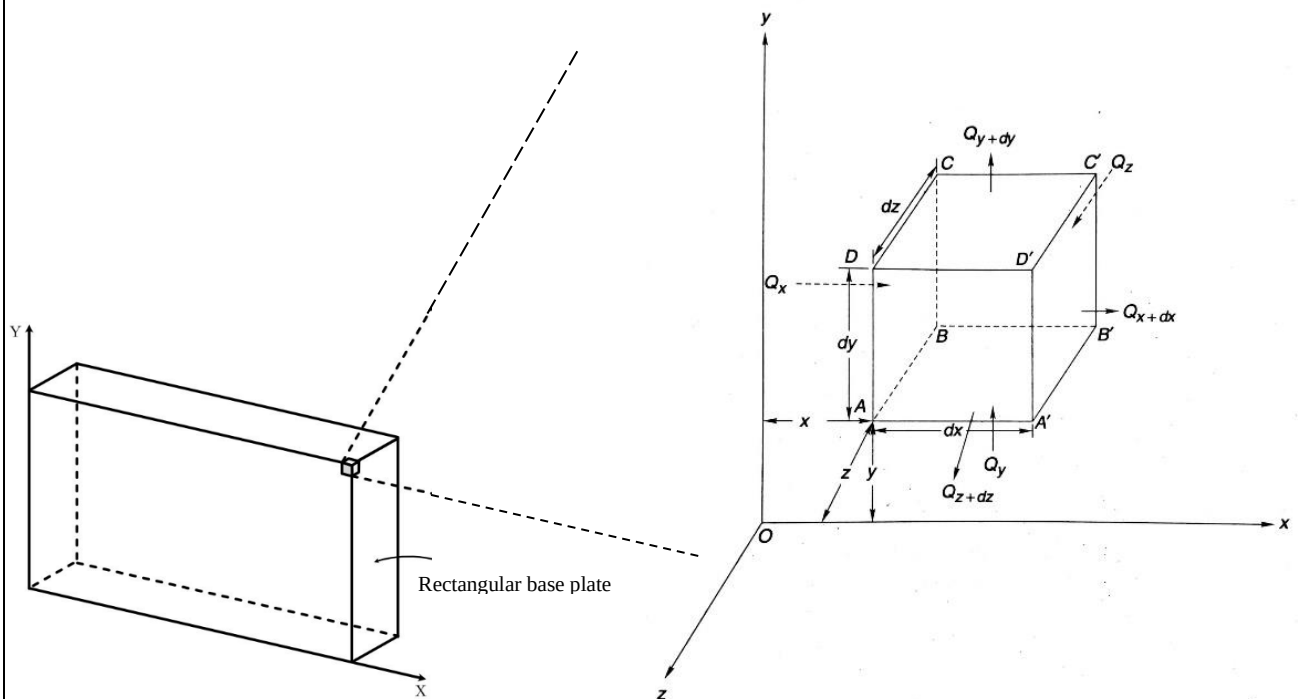


Fig 2.1.3.1 Differential element in cartesian coordinate system.

The last term of Eqn. (2.1.3.1) is very small because the flow work done by solids due to temperature.

The three terms, I, II and III of this equation are evaluated as follows:

At x (at face ABCD), Let q_x be the heat flux in x -direction and at $x + dx$ (at face A'B'C'D'), q_{x+dx} be the heat flux.

Then rate of heat flow into the element in x -direction through face ABCD according to Fourier's law is:

$$Q_x = -k_x \frac{\partial T}{\partial x} dy dz = q_x dy dz \text{ -----(2.1.3.2)}$$

where k_x is the thermal conductivity of material in x -direction and $\frac{\partial T}{\partial x}$ is the temperature gradient in x -direction.

The rate of heat flow out of the element in x -direction through the face at $x + dx$, A'B'C'D' according to Taylor's series is:

$$Q_{x+dx} = Q_x + \frac{\partial Q_x}{\partial x} \frac{dx}{1!} + \underbrace{\frac{\partial^2 Q_x}{\partial x^2} \frac{dx^2}{2!} + \dots}_{\text{Can be neglected}}$$

$$Q_{x+dx} = Q_x + \frac{\partial Q_x}{\partial x} dx = -k_x \frac{\partial T}{\partial x} dy \cdot dz + \frac{\partial}{\partial x} \left(-k_x \frac{\partial T}{\partial x} \right) dx dy dz \text{ -----(2.1.3.3)}$$

Then, the net rate of heat entering the element in x -direction is the difference between the entering and leaving heat flow rates, and is given by:

$$\left. \begin{aligned} Q_x - Q_{x+dx} &= \frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) dx dy dz \\ \text{Similarly,} \\ Q_y - Q_{y+dy} &= \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) dx dy dz \\ Q_z - Q_{z+dz} &= \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) dx dy dz \end{aligned} \right\} \text{ -----(2.1.3.4)}$$

The net heat conducted into the element $dx dy dz$ per unit time, term I in Eqn. (2.1.3.1), is:

$$I = \left[\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) \right] dx dy dz \text{ -----(2.1.3.5)}$$

Let $\dot{q}$ be the internal heat generation per unit time and per unit volume (W/m^3), the rate of energy generation in the element, term II in Eqn. (2.1.3.1), is

$$\text{II} = \dot{q} dx dy dz \text{ -----(2.1.3.6)}$$

The change in internal energy for the element per unit time, term III of Eqn. (2.1.3.1) is:

$$\begin{aligned} \text{III} &= mc_p \frac{\partial T}{\partial t} \text{ -----(2.1.3.7)} \\ &= \rho V c_p \frac{\partial T}{\partial t} \end{aligned}$$

where ρ , c_p are the density, specific heat and the volume of the material of the element.

$$\begin{aligned} &= \rho dx dy dz (c_p) \frac{\partial T}{\partial t} \\ \text{III} &= \left(\rho c_p \frac{\partial T}{\partial t} \right) dx dy dz \text{ -----(2.1.3.8)} \end{aligned}$$

Substitution of Eqns. (1.3.5), (1.3.6) and (1.3.8) into Eqn. (1.3.1) leads to the general three-dimensional equation for heat conduction:

$$\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t} \text{ -----(2.1.3.9)}$$

- ✓ The first three terms on the left-hand side are **the heat conduction terms** in the x-, y- and z-directions, respectively.
- ✓ The fourth term on the left-hand side is **the heat generation term** in the element of the material.
- ✓ The term on the right-hand side is **the rate of change of temperature term**.

Assume the materials can be considered isotropic for which $k_x = k_y = k_z = k = \text{constant}$, the general three-dimensional heat conduction equation becomes:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{\rho c_p}{k} \frac{\partial T}{\partial t} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \text{ -----(2.1.3.10)}$$

The quantity $\frac{k}{\rho c_p}$ is known as the **thermal diffusivity**, α of the material: It has got the units m^2/s (L^2/t).

The thermal diffusivity:

The physical significance of thermal diffusivity is that it tells us how fast heat is propagated or it diffuses through a material during changes of temperature with time. The larger the thermal diffusivity, the shorter is the time required for the applied heat to penetrate deeper into the solid. (*further discuss later)

Special forms of the heat conduction equation

Let us now examine some special cases of Eqn. (2.1.3.9).

(i) Uniform thermal conductivity

When the thermal conductivity of the material is independent of position and temperature, Eqn. (2.1.3.9) takes the form:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \text{ -----(2.1.3.11)}$$

or

$$\nabla^2 T + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \text{ -----(2.1.3.12)}$$

where $\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$ is called the **Laplacian (operator)** of T .

(ii) Steady state conditions

The temperature at any point in the material does not change with time, i.e., $\frac{\partial T}{\partial t} = 0$. The heat conduction equation for this case becomes.

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = 0 \text{ -----(2.1.3.13)}$$

or

$$\nabla^2 T + \frac{\dot{q}}{k} = 0 \text{ (Poisson equation)}$$

(iii) No Heat Sources

In the absence of any heat generation or release of energy within the body, Eqn. (2.10) becomes:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \text{ (Diffusion equation) -----(2.1.3.13)}$$

If in addition, the process is in a steady state, then the heat conduction equation becomes:

$$\begin{aligned} \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} &= 0 \\ \nabla^2 T &= 0 \text{ (Laplace equation)} \end{aligned} \text{ -----(2.1.3.14)}$$

(iv). One-Dimensional Heat Conduction

If the temperature varies only in the x -direction, then the Laplacian operator $\nabla^2 T$ takes the form $\frac{d^2 T}{dx^2}$ in all these equations, e.g., the conduction Eqn. (2.1.3.14) for a steady case with no heat generation is:

$$\frac{d^2 T}{dx^2} = 0 \text{ -----(2.1.3.15)}$$

2.1.4. INITIAL AND BOUNDARY CONDITIONS

The temperature distribution in a medium of given form and size can be determined by solving an appropriate differential equation subject to given constraints, called the initial and boundary conditions.

- ✓ The initial conditions describe the temperature distribution in a medium at the initial moment of time.

In general, these can be expressed as,

$$\text{at } t = 0, \quad T = T(x, y, z) \text{ -----(2.1.4.1)}$$

A simple but typical form of Eqn. (2.26) for a uniform initial temperature distribution is,

$$\text{at } t = 0, \quad T = T_0 = \text{constant} \text{ -----(2.1.4.2)}$$

- ✓ The boundary conditions specify the temperature or the heat flow at the surface of the body. Boundary conditions may be prescribed in several ways.

(i). Boundary condition of the first kind (prescribed surface temperature)

A boundary condition of the first kind (prescribed surface temperature) is a type of boundary condition that specifies the value of the temperature at the boundary of the domain. For example, if we have a heat sink attached to a component, we can specify the temperature of the component as a boundary condition of the first kind for the heat sink. This means that the heat sink and the component are in thermal equilibrium at the interface.

A boundary condition of the first kind is also known as a **Dirichlet boundary condition**, or a **fixed boundary condition**. It is one of the most common types of boundary conditions used in heat transfer problems.

The temperature distribution, T_s , is given at a boundary surface for each moment of time.

$$T_s = T(x, y, z, t) \text{ -----(2.1.4.3)}$$

A typical example of boundary condition of the first kind for a slab is shown in Fig. 2.1.4.1

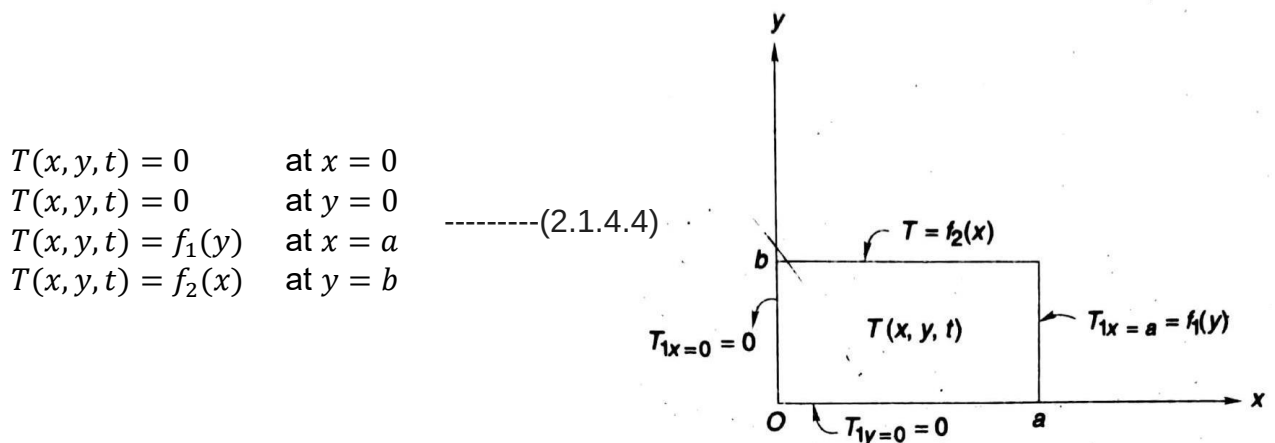
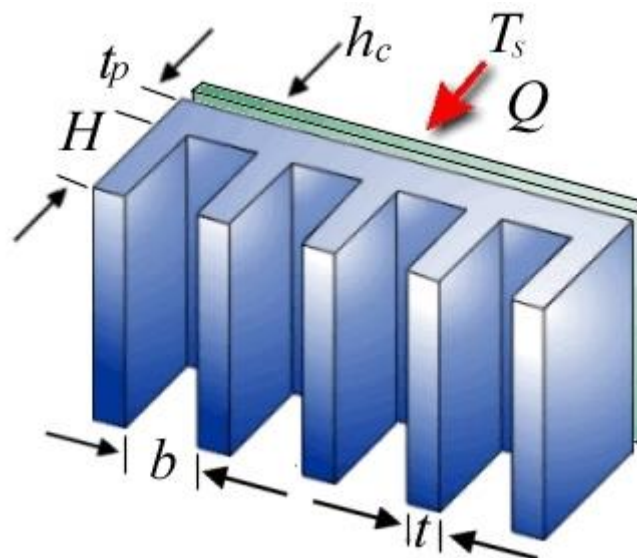


Fig. 2.1.4.1 Boundary Condition of the First Kind

A heat sink is a device that cools down a component by transferring heat to a fluid medium, such as air or water. A boundary condition of the first kind specifies the temperature of the component or the fluid at the interface with the heat sink. This affects how much heat flows from the component to the heat sink and from the heat sink to the fluid.



For example, if we have a rectangular heat sink attached to a component, we can use a boundary condition of the first kind to specify the temperature of the component as T_c at the interface with the heat sink. Then, we can use the general conduction equation to solve for the temperature distribution and heat transfer rate in the heat sink. We can also use a boundary condition of the first kind to specify the temperature of the fluid as T_f at the interface with the heat sink. This affects how much heat is removed from the heat sink by convection or radiation.

(ii). Boundary condition of the second kind (prescribed heat flux)

A boundary condition of the second kind specifies the heat flux or the rate of heat transfer per unit area at the boundary of the domain. This affects how much heat is delivered or removed from the heat sink by conduction, convection, or radiation.

In this case, the heat flux at a boundary is prescribed, this can be expressed as (see Fig. 2.1.4.2).

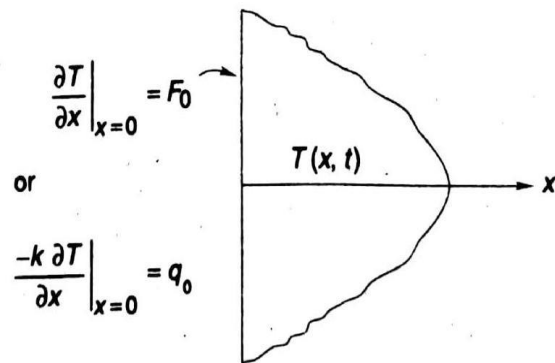


Fig. 2.1.4.2 Boundary Condition of the Second Kind

$$-k \frac{\partial T(x,t)}{\partial x} = q_0 \quad \text{at } x = 0$$

or -----(2.1.4.5)

$$+\frac{\partial T}{\partial x} = -\frac{q_0}{k} = F_0 \quad \text{at } x = 0$$

$\frac{\partial T}{\partial x} = 0$ at $x = 0$ describes an insulated or adiabatic boundary. This condition can also exist at the plane of symmetry.

For example, if we have a rectangular heat sink attached to a component, we can use a boundary condition of the second kind to specify the heat flux from the component as q at the interface with the heat sink. Then, we can use the general conduction equation to solve for the temperature distribution and heat transfer rate in the heat sink. we can also use a boundary condition of the second kind to specify the heat flux from the fluid as $h(T_f - T)$ at the interface with the heat sink, where h is the convection co-efficient, T_f is the fluid temperature and T is the heat sink temperature. This affects how much heat is removed from the heat sink by convection.

(iii). Boundary condition of the third kind (convective condition)

A boundary condition of the third kind specifies the relation between the heat flux and the temperature difference at the boundary of the domain. This accounts for the effects of convection and radiation on the heat transfer rate and temperature distribution in the heat sink.

One commonly encountered situation at a solid boundary is the equality between heat transfer to the surface by conduction and that leaving the surface by convection as shown in Fig. 2.1.4.3

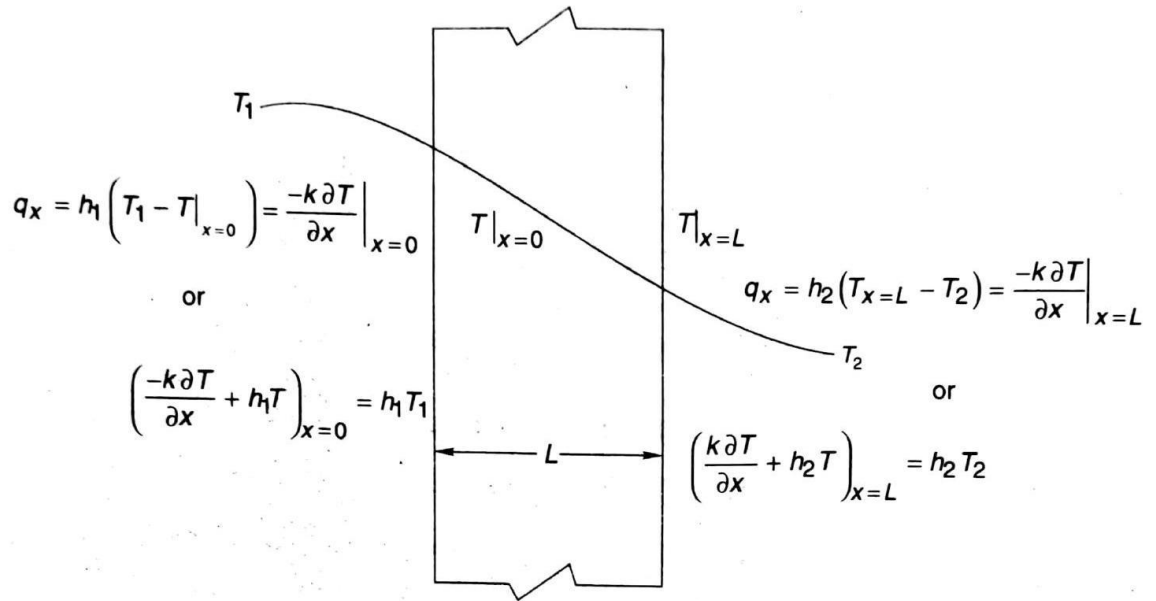


Fig. 2.1.4.3 Boundary Condition of the third kind

at $x = 0$,

$$h_1 (T_1 - T|_{x=0}) = -k \frac{\partial T}{\partial x} \Big|_{x=0}$$

Or

----- (2.1.4.6)

$$\left(k \frac{\partial T}{\partial x} + h_1 T \right)_{x=0} = h_1 T_1 = F_1$$

Similarly,

$$\text{at } x = L, \quad -k \frac{\partial T}{\partial x} \Big|_{x=L} = h_2 (T|_{x=L} - T_2)$$

Or

----- (2.1.4.7)

$$\left(k \frac{\partial T}{\partial x} + h_2 T \right)_{x=L} = h_2 T_2 = F_2$$

The boundary conditions described by Eqns. (2.1.4.6) and (2.1.4.7) are called the boundary conditions of the third kind, which can be expressed in a compact form as:

$$\left(k \frac{\partial T}{\partial n_i} + h_i T\right)_{\text{boundary}} = F_i \quad \text{-----}(2.1.4.8)$$

where h_i is a specific coefficient, F_i is a prescribed function, and $\frac{\partial T}{\partial n_i}$ is the normal derivative of the boundary surface in **the direction of the outward normal**. For the case of the slab shown in Fig. 2.1.4.3 the outward normal at $x = L$ is in the positive x -direction whereas at $x = 0$, it is in the negative x -direction. Eqns. (2.1.4.6) and (2.1.4.7) can be obtained from the general Eqn. (2.1.4.8) by considering this sign convention.

(iv). Boundary condition of the fourth kind

A boundary condition of the fourth kind is a type of boundary condition that involves a mixed form of heat transfer at the boundary of the domain. For example, if we have a heat sink in contact with a fluid medium, we can use a boundary condition of the fourth kind to model the heat transfer between the heat sink and the fluid as a combination of conduction, convection, and radiation.

Mathematically, it can be written as:

$$q = h(T_f - T) + \epsilon\sigma(T_f^4 - T^4) - k \frac{dT}{dn} \quad \text{-----}(2.1.4.9)$$

where q is the heat flux, h is the convection coefficient, T_f is the fluid temperature, T is the heat sink temperature, ϵ is the emissivity, σ is the Stefan-Boltzmann constant, k is the thermal conductivity, and $\frac{dT}{dn}$ is the temperature gradient normal to the boundary. The first term on the right-hand side represents the heat transfer by convection, the second term represents the heat transfer by radiation, and the third term represents the heat transfer by conduction.

A boundary condition of the fourth kind is also known as a Stefan boundary condition or a mixed boundary condition. It is one of the most realistic types of boundary conditions used in heat transfer scenarios. It can be applied to ordinary or partial differential equations that describe heat conduction, convection, or radiation.

This boundary condition describes the process of conduction between two bodies in perfect contact. Since the surfaces in contact are at the same temperature, the heat fluxes through them are equal, (see Fig. 2.1.4.4).

$$k_1 \left(\frac{\partial T_1}{\partial n}\right)_i = k_2 \left(\frac{\partial T_2}{\partial n}\right)_i \quad \text{-----}(2.1.4.10)$$

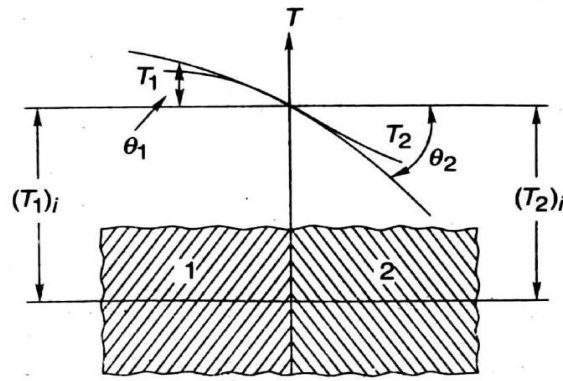


Fig. 2.1.4.4 Boundary Condition of the Fourth Kind

An obvious inference from this condition is that the ratio of slopes of the tangents to the temperature curves at the points of contact of two bodies is a constant.

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{k_2}{k_1} = \text{constant} \text{ -----(2.1.4.11)}$$

2.1.5. ANISOTROPIC MATERIALS FOR HEAT SINK

Anisotropic materials have different thermal conductivity properties in different directions. This can be beneficial in heat sinks because it allows the heat to be transferred more efficiently in the desired direction.

This can be beneficial in heat sinks because it allows the heat to be transferred more efficiently in the desired direction. For example, if we have a heat sink made of an anisotropic material with high thermal conductivity in the vertical direction and low thermal conductivity in the horizontal direction, we can expect that the heat will flow mainly from the bottom to the top of the heat sink, where it can be dissipated by convection or radiation to the fluid medium. This way, the heat sink can maintain a low temperature difference between the component and the fluid and achieve a high heat transfer rate.

However, anisotropic materials can also be more difficult to manufacture than isotropic materials. This is because the manufacturing process needs to consider the anisotropy of the material.

In general, the choice of whether to use an isotropic or anisotropic material in a heat sink depends on the specific application. If the heat sink needs to transfer heat efficiently in a specific direction, then an anisotropic material may be a better choice. However, if the heat sink needs to be easy to manufacture, then an isotropic material may be a better choice.

Here are some examples of isotropic materials that are commonly used in heat sinks:

- Aluminum
- Copper
- Stainless steel

Here are some examples of anisotropic materials that are sometimes used in heat sinks:

- Graphite
- Carbon fiber
- Composite materials

Here are some additional factors to consider when choosing a material for a heat sink:

- **Thermal conductivity:** The thermal conductivity of the material determines how well it will transfer heat.
- **Density:** The density of the material determines how much heat it can store.
- **Cost:** The cost of the material is also a key factor to consider.

2.1.6. THERMAL DIFFUSIVITY (α) FOR HEAT SINK

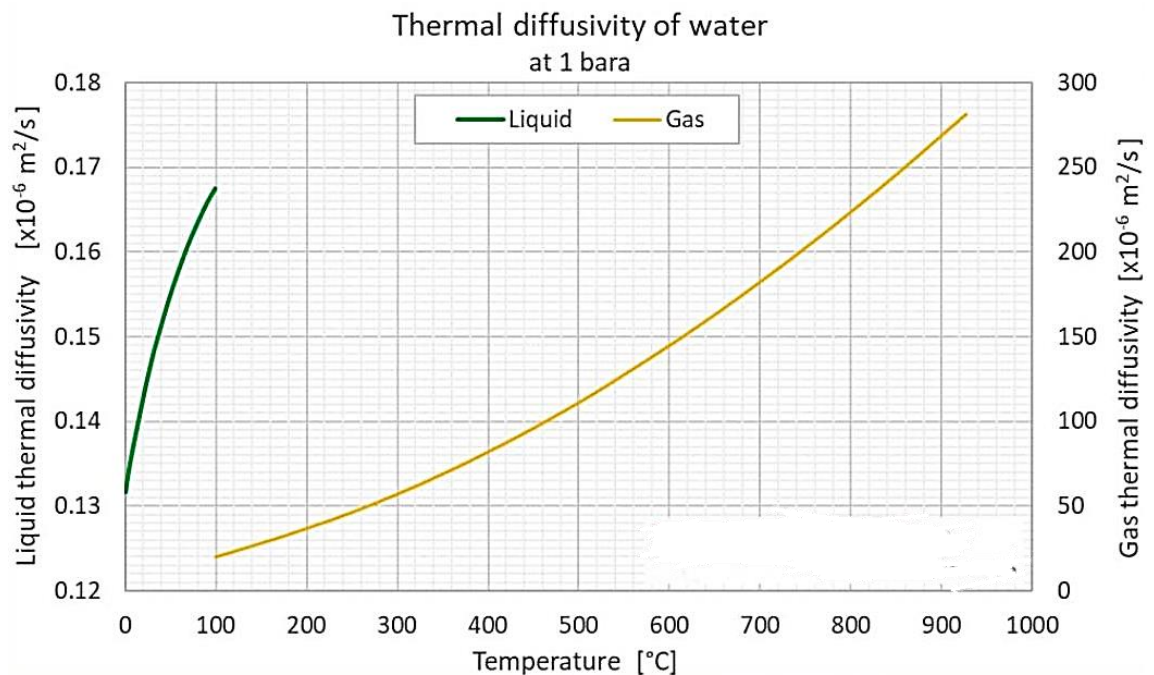
$$\alpha = \frac{k}{\rho C_p} \text{ -----(2.1.6.1)}$$

where α is the thermal diffusivity, k is the thermal conductivity, ρ is the density, and c_p is the specific heat capacity. The thermal diffusivity has units of m^2/s and depends on the material and temperature.

- Thermal diffusivity is a measure of the transient thermal response of a material to a change in temperature.
- The larger the value of α , the faster will the heat diffuse through the material and its temperature will change with time.
- This will result either due to a high value of conductivity k or a low value of ρ, C_p .
- Thermal diffusivity is a convenient collection of physical properties for transient solutions of the heat equation.
- Recollect about 'kinematic viscosity' ($\nu = \mu/\rho$), which is also called as 'momentum diffusivity', and D_{AB} .

Material	Thermal Diffusivity (cm ² /s) at 300 K
Copper	1.15
Aluminum	0.97
Stainless Steel (304)	0.042
Silicon Dioxide (Polycrystalline)	0.0083
Water	0.0014
Polyvinyl Chloride (PVC)	0.0008
Alcohol	0.0007
Air	0.19

- Metals and gases have relatively high value of thermal diffusivity and their response to temperature changes is quite rapid.
- The nonmetallic solids and liquids respond slowly to temperature changes because of their relatively small value of thermal diffusivity.



A heat sink is a device that cools down a component by transferring heat to a fluid medium, such as air or water. A heat sink is required to have a very high thermal diffusivity that enables the quick transport of the heat. If slow heat transfer were to occur, the area accepting the heat would heat up and not permit as much heat flow per unit time. This would result in a high temperature difference between the component and the fluid, and a low heat transfer rate.

The general heat conduction equation for a steady-state, one-dimensional problem with no heat generation is:

$$\frac{d^2T}{dx^2} = 0 \text{ -----(2.1.6.2)}$$

where T is the temperature and x is the distance along the heat sink. The solution of this equation is:

$$T = C_1x + C_2 \text{ -----(2.1.6.3)}$$

where C_1 and C_2 are constants that depend on the boundary conditions. The heat flux at any point along the heat sink is given by Fourier's law:

$$q_x = -k \frac{dT}{dx} = -kC_1 \text{ -----(2.1.6.4)}$$

where q_x is the heat flux in the x-direction. The total rate of heat transfer through the heat sink is given by:

$$Q = q_xA = -kC_1A \text{ -----(2.1.6.5)}$$

where A is the cross-sectional area of the heat sink. From these equations, we can see that the rate of heat transfer depends on the thermal conductivity of the material, as well as the temperature gradient along the heat sink. The thermal diffusivity affects how fast the temperature gradient changes along the heat sink. A high thermal diffusivity means that the temperature gradient is low and uniform, which leads to a high rate of heat transfer. A low thermal diffusivity means that the temperature gradient is high and non-uniform, which leads to a low rate of heat transfer.

2.2. CONVECTION PROCESS IN A HEAT SINK

2.2.1 HEAT TRANSFER FROM EXTENDED SURFACES – THE FINS

The paragraph discusses the challenge of heat transfer between a gas and a liquid due to the lower thermal conductivities of gases compared to liquids. This leads to a smaller heat transfer coefficient on the gas side, resulting in the gas-side 'film'* presenting most of the thermal resistance. Consequently, a larger heat transfer area becomes necessary.

To address this issue, a practical solution involves increasing the surface area available for heat transfer on the gas side. This can be achieved by attaching fins, such as rectangular metal strips or annular rings, to the heat transfer surface. Fins serve to enhance the surface area and are commonly referred to as extended surfaces.

However, when using a finned pipe with a hot fluid, the temperature of the fin's surface is lower than that of the pipe wall. This temperature difference reduces the driving force for heat transfer from the fin's surface. To analyze the effectiveness of rectangular fins in enhancing heat transfer rates, a theoretical analysis is provided.

The analysis considers a thin rectangular plate that extends a distance ' l ' from a wall. This fin has a length ' l ', a thickness ' w ', and a breadth ' b '. The wall temperature is ' T_w ' (referred to as the 'base temperature' of the fin), and the ambient temperature is ' T_o '. Heat loss to the ambient occurs from both surfaces of the fin (top and bottom). As the fin thickness is small, it is assumed that the temperature is uniform across any transverse section of the fin. The fin's temperature varies only longitudinally, and heat transfer predominantly occurs through the fin via conduction in the same direction.

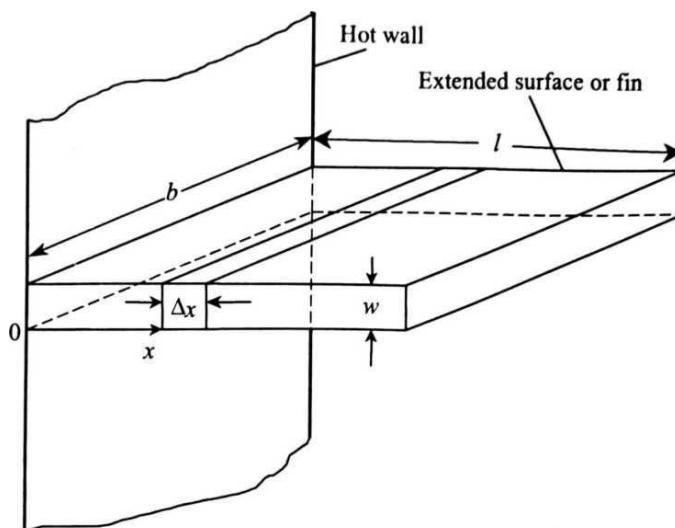


Fig. 2.2.1.1 – Heat balance in a rectangular fin

* - In the context of heat transfer, the term "film" refers to a thin layer or boundary that forms on the surface of an object, such as a solid wall or a pipe, when it meets a fluid (liquid or gas). This layer is often referred to as a "boundary layer" or a "fluid film."

Let us now consider a thin section of thickness Δx of the fin at a distance x from the base. For the breadth b of the fin, the available area of heat conduction is bw and the area of heat loss by convection from the exposed fin surface to the ambient is $p\Delta x$ [where p is the fin perimeter $= 2(b + w)$]. In order to make a heat balance over the small element, we identify the following heat input and output terms.

$$\begin{aligned} \text{Rate of heat input at } x &= bw q_x|_x \\ \text{Rate of heat output at } x+\Delta x &= bw q_x|_{x+\Delta x} \\ \text{Rate of heat lost from the exposed surfaces} &= p\Delta x h(T - T_0) \\ \text{Rate of heat accumulation (at steady state)} &= 0 \end{aligned}$$

where T is the local temperature of the fin and h is the surface heat transfer coefficient. Making a steady state heat balance, we have

$$bw q_x|_x - bw q_x|_{x+\Delta x} - p\Delta x h(T - T_0) = 0$$

Dividing by $bw\Delta x$ throughout and taking *limit* $x \rightarrow 0$, we get

$$-\frac{d}{dx}(q_x) = \frac{ph}{bw}(T - T_0)$$

Putting $q_x = -kdT/dx$, and defining $\bar{T} = T - T_0$ we have

$$\frac{d^2\bar{T}}{dx^2} = \frac{ph}{kbw}\bar{T} = \xi^2\bar{T}$$

$$\text{Where } \xi = \sqrt{\frac{ph}{kbw}} \quad \text{----- (eq. 2.2.1.1)}$$

To solve above equation for the temperature distribution in the fin we need to write down the boundary conditions at $x = 0$ and $x = l$.

$$\text{At } x=0, T=T_w, \text{ i.e. } \bar{T} = T_w - T_0 \quad \text{----- (eq. 2.2.1.2)}$$

$$\text{At } x=l, -k\frac{dT}{dx} = h(T - T_0), \text{ i.e. } -k\frac{d\bar{T}}{dx} = h\bar{T} \quad \text{----- (eq. 2.2.1.3)}$$

Eq. 2.2.1.3 is a *convective boundary condition*. It implies that the rate at which heat is conducted from inside the solid to the boundary must be equal to the rate at which it is transported to the ambient medium by convection.

Solution to Eq. 2.2.1.2 is

$$\begin{aligned} \bar{T} &= C_1 \cosh(\xi x) + C_2 \sinh(\xi x) \\ &\text{(eq. 2.2.1.4)} \end{aligned}$$

where C_1 and C_2 are integration constants.

Using the boundary condition given by Eq. 2.2.1.2, we get

$$C_1 = T_w - T_0$$

(eq. 2.2.1.5)

Differentiating $\bar{T}$ in Eq. 2.2.1.4 and using Eq. 2.2.1.3, we get

$$-k[\xi C_1 \sinh(\xi l) + \xi C_2 \cosh(\xi l)] = h[C_1 \cosh(\xi l) + C_2 \sinh(\xi l)]$$

Or

$$C_2 = -\frac{C_1 \left[\frac{h}{k\xi} \cosh(\xi l) + \sinh(\xi l) \right]}{\cosh(\xi l) + \sinh(\xi l)}$$

(eq. 2.2.1.6)

Substitution of the relations for C_1 and C_2 in Eq. 2.2.1.4 and simplification lead us to the final solution for the *dimensionless fin temperature* given below.

$$\frac{\bar{T}}{T_w - T_0} = \frac{T - T_0}{T_w - T_0} = \frac{\cosh \xi(l - x) + \frac{h}{k\xi} \sinh \xi(l - x)}{\cosh(\xi l) + \frac{h}{k\xi} \sinh \xi l}$$

(eq. 2.2.1.7)

It may be seen from the solution [Eq. 2.2.1.7] that the temperature decreases with distance x from the base. So the surface heat flux from the fin also decreases in the longitudinal direction because of a smaller temperature driving force. It is easy to visualize that if the thermal conductivity k of the material of the fin is extremely high, its thermal resistance will be negligibly small and the temperature will remain almost uniform throughout at T_w , which is the wall or base temperature. The maximum rate of heat transfer can be achieved under this condition, which is rather an ideal condition. It is, therefore, interesting, and useful to calculate the actual rate of heat transfer and also its ratio to the maximum rate of heat transfer.

The actual rate of heat loss from the fin must be equal to the rate of heat input by conduction at the fin base from the wall at steady state. Therefore, we have

$$Q = (\text{cross-sectional area}) (\text{heat flux at the fin base}) = (bw) [-k(dT/dx)]_{x=0}$$

The temperature gradient dT/dx can be determined from Eq. 2.2.1.7. Therefore, we get

$$Q = bw(T_w - T_0)(-k) \left[\frac{-\xi \sinh \xi(l - x) - \frac{h}{k\xi} \cosh \xi(l - x)}{\cosh \xi l + \frac{h}{k\xi} \sinh \xi l} \right]_{x=0}$$

(eq. 2.2.1.8)

If the fin temperature remains uniform at T_w (i.e. the base temperature), the rate of heat loss is the maximum

$$Q_{\max} = (\text{total area})(\text{heat transfer coefficient})(\text{driving force}) = (pl + bw)(h)(T_w - T_0)$$

(eq. 2.2.1.9)

The fin efficiency η is defined as

$$\eta = \frac{\text{Actual rate of heat transfer}}{\text{Calculated rate of heat transfer if the temperature of the fin remains uniform at the base value, } T_w}$$

Therefore, from Eqs. (2.2.1.8) and (2.2.1.9) we get

$$\eta = \frac{Q}{Q_{\max}} = \frac{bwk\xi}{h(pl + bw)} \frac{\sinh \xi l + \frac{h}{k\xi} \cosh \xi l}{\cosh \xi l + \frac{h}{k\xi} \sinh \xi l}$$

(eq. 2.2.1.10)

A simpler but nevertheless accurate expression for heat loss can be obtained by using an alternative form of the boundary condition Eq. 2.2.1.3. If we assume that heat loss from the edge of the fin (at $x = l$) is negligibly small, we can write

$$x=l, \quad -k \frac{dT}{dx} = 0 \text{ ----- (eq. 2.2.1.11)}$$

Solution of Eq. 2.2.1.1 subject to boundary conditions (2.2.1.2) and (2.2.1.11) can be obtained as,

$$\frac{T - T_0}{T_w - T_0} = \frac{\cosh [\xi(l - x)]}{\cosh \xi l}$$

(eq. 2.2.1.12)

which is the longitudinal temperature distribution in the fin. The rate of heat loss from both the surfaces (neglecting that from the edge) is

$$Q = (bw) - \left[k \frac{dT}{dx} \right]_{x=0} = bwk\xi \frac{(T_w - T_0) \sinh \xi l}{\cosh \xi l}$$

(eq. 2.2.1.13)

The fin efficiency is given by,

$$\eta = \frac{Q}{Q_{max}} = \frac{b w k \xi}{h p l (T_w - T_0)} \frac{(T_w - T_0) \sinh \xi l}{\cosh \xi l} = \frac{1}{\xi l} \tanh \xi l \quad \because \xi^2 = \frac{p h}{k b w}$$

(eq. 2.2.1.14)

If the fin thickness w is small, $p = 2(b + w) \approx 2b$, then $\xi^2 = 2h/kw$. It is easy to check the accuracy of Eq. 1.2.14 for a physically realistic situation. If we take $l = 5$ cm, $w = 4$ mm, $b = 1$ m, $k = 160$ W/m °C, $h = 50$ W/m °C, then $\xi = [ph/kbw]^{1/2} = 12.5$ m⁻¹, and $\xi l = 0.625$. From Eq. 2.2.1.10, $\eta = 0.88$; and from Eq. 14, $\eta = 0.887$. These values of η are very close to each other. Equation 2.2.1.14 is good enough for practical purposes.

(This calculation for heat sink can be further simplified by assuming the fin is infinitely long. So then we can take that temperature at the end of the fin equals to ambient temperature (T_0). Then,

limit $x \rightarrow 0$,

$$\bar{T} = T - T_0 = T_0 - T_0 = 0$$

We can take that, $\bar{T} = C_1 e^{\xi x} + C_2 e^{-\xi x}$ and $C_1 = 0$

When $x = 0$,

$$\bar{T} = T_w - T_0$$

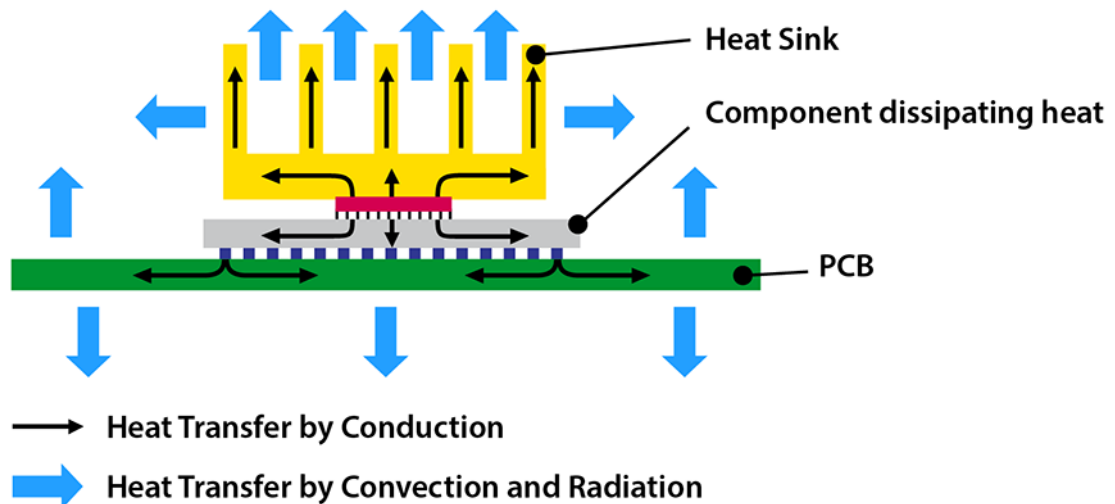
Then, $C_2 = T_w - T_0$

Therefore,

$$\bar{T} = (T_w - T_0) e^{-\xi x}$$

)

The dissipation of heat transferred through conduction to the environment is accomplished through the process of convection. However, there are further important considerations when it comes to convection, particularly in the design of heat sinks. Convection methods play a critical role in heat sink design to ensure effective heat dissipation. Now it's better to have a deep understanding about what convection is.



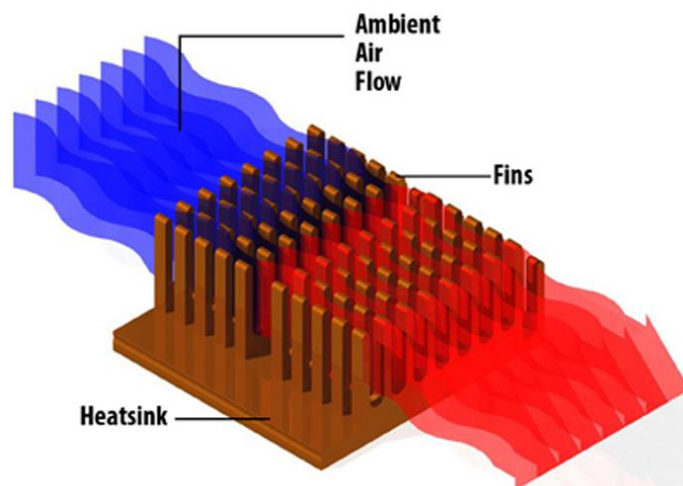
2.2.2 IMPORTANCE OF CONVECTION WHEN ANALYZING HEAT SINKS

Heat sinks, which are commonly used in electronic devices to dissipate heat, can be categorized as active or passive based on their cooling mechanisms.

Passive heat sinks rely on natural convection for heat transfer. Natural convection occurs when heat is transferred due to the density difference caused by temperature gradients. In passive heat sinks, the design typically incorporates fins that increase the surface area for heat dissipation. As the surrounding air gets heated by the heat sink, it becomes less dense and rises, creating a natural flow that aids in heat transfer. Passive heat sinks are commonly used in applications where the heat dissipation requirements are relatively low or in situations where power consumption needs to be minimized.

Active heat sinks, on the other hand, utilize forced convection for enhanced heat transfer. Forced convection involves using external means, such as fans or pumps, to induce airflow and improve heat dissipation. Active heat sinks typically have a fan or blower attached, which forcefully circulates air over the heat sink, increasing the heat transfer rate. This approach is used when higher heat dissipation is required or when dealing with heat-intensive applications like high-performance computing or industrial equipment.

The choice between forced and natural convection for heat sinks depends on factors such as the heat load, available space, power consumption constraints, and desired cooling efficiency. Forced convection offers more efficient heat dissipation, but it comes with the trade-off of increased power consumption and noise due to the need for fans or blowers. Natural convection, on the other hand, is quieter and requires no additional power, but it may be less effective in dissipating high heat loads. Therefore, the selection of convection method is based on balancing these factors to achieve optimal cooling performance for the specific application.



2.2.3 IMPORTANT THERMODYNAMIC CORRELATIONS

Thermodynamic correlations are important in determining the thermodynamics of a heat sink because they provide a way to predict the heat transfer rate between the heat sink and the surrounding environment. This is important because the heat transfer rate determines how efficiently the heat sink can remove heat from the electronic device it is cooling.

There are several different thermodynamic correlations that can be used to predict the heat transfer rate of a heat sink. These correlations are based on experimental data and theoretical models of heat transfer. The choice of which correlation to use depends on the specific design of the heat sink and the surrounding environment.

In general, thermodynamic correlations are more accurate for heat sinks with simple geometries. For heat sinks with complex geometries, it may be necessary to use computational fluid dynamics (CFD) to accurately predict the heat transfer rate.

2.2.3.1. NUSSELT NUMBER

The Nusselt number is a dimensionless number that is used to quantify the rate of heat transfer between a solid surface and a fluid. It is defined as the ratio of the convective heat transfer coefficient to the thermal conductivity of the fluid. The Nusselt number is important in determining the thermodynamics of a heat sink because it is a measure of how efficiently the heat sink can transfer heat away from the electronic device it is cooling.

$$\text{Nusselt number} = \frac{\text{Wall temperature gradient}}{\text{Temperature gradient across the fluid in the pipe}}$$

$$Nu = \frac{hl}{k}$$

h - heat transfer coefficient ($\text{Wm}^{-2}\text{K}^{-1}$)

k - Thermal conductivity ($\text{Wm}^{-1}\text{K}^{-1}$)

A higher Nusselt number indicates that the heat sink is more efficient at transferring heat. This is because a higher Nusselt number means that the convective heat transfer coefficient is higher, which means that more heat is transferred from the solid surface to the fluid.

2.2.3.2. REYNOLDS NUMBER

The Reynolds number is a dimensionless quantity that is used to characterize the flow of a fluid. It is defined as the ratio of inertial forces to viscous forces. The Reynolds number is important in determining the type of flow that occurs in a heat sink. For example, if the Reynolds number is less than 2,000, the flow is laminar. If the Reynolds number is greater than 2,000, the flow is turbulent.

The type of flow that occurs in a heat sink affects the heat transfer rate. In laminar flow, the heat transfer rate is relatively low. In turbulent flow, the heat transfer rate is much higher. Therefore, the Reynolds number is an important factor in determining the thermodynamics of a heat sink.

$$\text{Reynolds number} = \frac{\text{Inertial force}}{\text{Viscous force}}$$

$$Re = \frac{IV\rho}{\mu}$$

V - Average fluid velocity (ms^{-1})

ρ - Density (kgm^{-3})

μ - Dynamic viscosity ($\text{kgm}^{-1}\text{s}^{-1}$)

Laminar flow of fluid converts to a turbulent flow at critical Reynolds number.

2.2.3.3. GRASHOF NUMBER

Grashof's number is a dimensionless quantity that is used to characterize the buoyancy-driven convection heat transfer. It is defined as the ratio of the buoyancy force to the viscous force. The Grashof number is important in determining the type of convection that occurs in a heat sink. For example, if the Grashof number is greater than 100, the convection is turbulent. If the Grashof number is less than 100, the convection is laminar.

The type of convection that occurs in a heat sink affects the heat transfer rate. In turbulent convection, the heat transfer rate is much higher than in laminar convection. Therefore, the Grashof number is an important factor in determining the thermodynamics of a heat sink.

$$\text{Grashof number} = \frac{\text{Buoyancy force}}{\text{Viscous force}}$$

$$Gr = \frac{g\beta L^3(T_w - T_\infty)}{\nu^2} = \frac{g\beta L^3 \rho^2(T_w - T_\infty)}{\mu^2}$$

β - Thermal Expansion Coefficient (K^{-1})

g - Acceleration due to Earth's gravity (ms^{-2})

ν - Kinematic Viscosity (m^2s^{-1})

μ - Dynamic Viscosity (Pas)

T_w - Wall Temperature

T_∞ - Bulk Temperature

2.2.3.4. PRANDTL NUMBER

The Prandtl number is a dimensionless quantity that is used to characterize the ratio of momentum diffusivity to thermal diffusivity. It is a measure of how well a fluid can transfer momentum compared to how well it can transfer heat.

The Prandtl number is important in determining the type of convection that occurs in a heat sink. For example, if the Prandtl number is less than 1, the convection is laminar. If the Prandtl number is greater than 1, the convection is turbulent.

$$\text{Prandtl number} = \frac{\text{Momentum Diffusivity}}{\text{Thermal Diffusivity}}$$

$$\text{Pr} = \frac{\nu}{\alpha} = \frac{c_p \mu}{k}$$

c_p - Specific Heat Capacity ($\text{Wskg}^{-1}\text{K}^{-1}$)

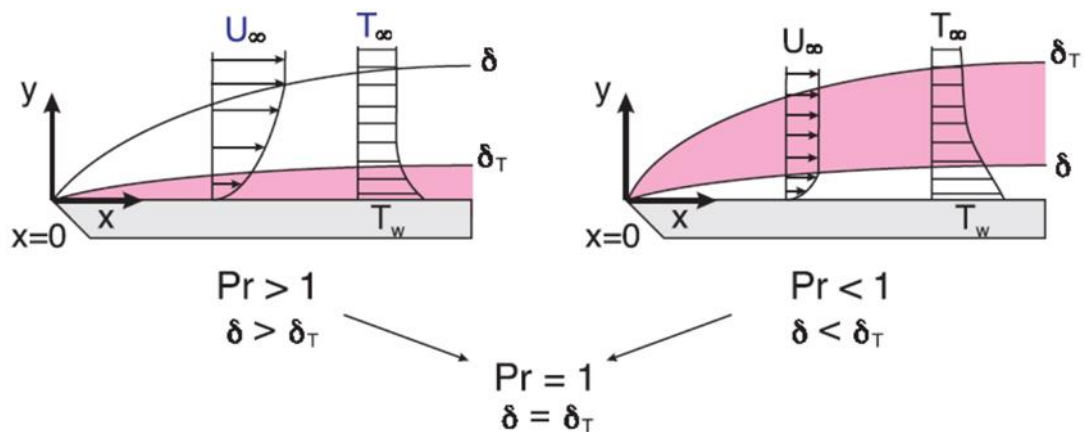
ν - Momentum Diffusivity

α - Thermal Diffusivity

Thermal diffusivity is a measure of how quickly heat can spread through a material. It is defined as the thermal conductivity of a material divided by its density and specific heat capacity. The higher the thermal diffusivity, the faster heat can spread through the material.

Momentum diffusivity is a measure of how quickly momentum can be transferred through a fluid. It is defined as the kinematic viscosity of a fluid divided by its density. The higher the momentum diffusivity, the faster momentum can be transferred through the fluid.

Prandtl number is also important in determining boundary layer (More discussions will be done on boundary layers later.)



2.2.3.5. RAYLEIGH NUMBER

The Rayleigh number is important because it determines whether or not natural convection will occur in a fluid. If the Rayleigh number is less than a critical value, then natural convection will not occur. However, if the Rayleigh number is greater than the critical value, then natural convection will occur and the fluid will start to circulate.

The critical Rayleigh number depends on the geometry of the system and the properties of the fluid. However, in general, the critical Rayleigh number is higher for fluids with a high viscosity and a low thermal diffusivity.

The Rayleigh number is a useful tool for predicting the onset of natural convection in a variety of systems. For example, the Rayleigh number can be used to predict the onset of convection in a heat sink or in the atmosphere.

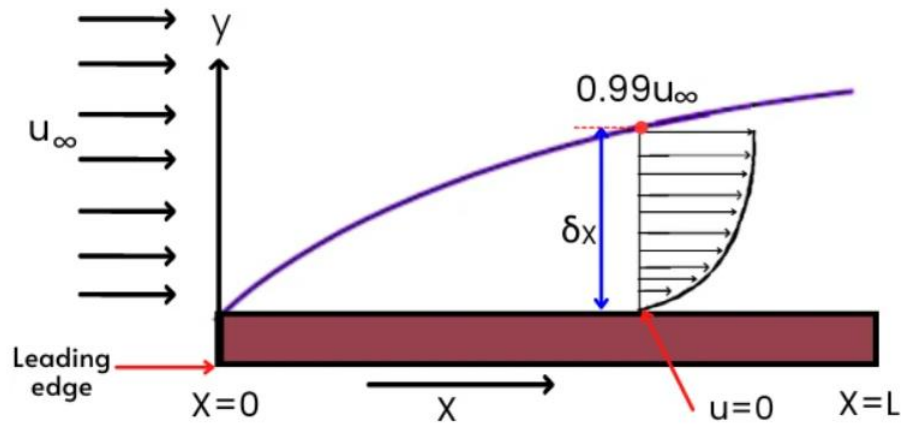
$$\text{Rayleigh Number} = (\text{Grashof Number}) \times (\text{Prandtl Number})$$

2.2.4. DETERMINING BOUNDARY LAYER FOR FLOW OF FLUID OVER A FLAT PLATE AND PRESSURE DROP CALCULATIONS

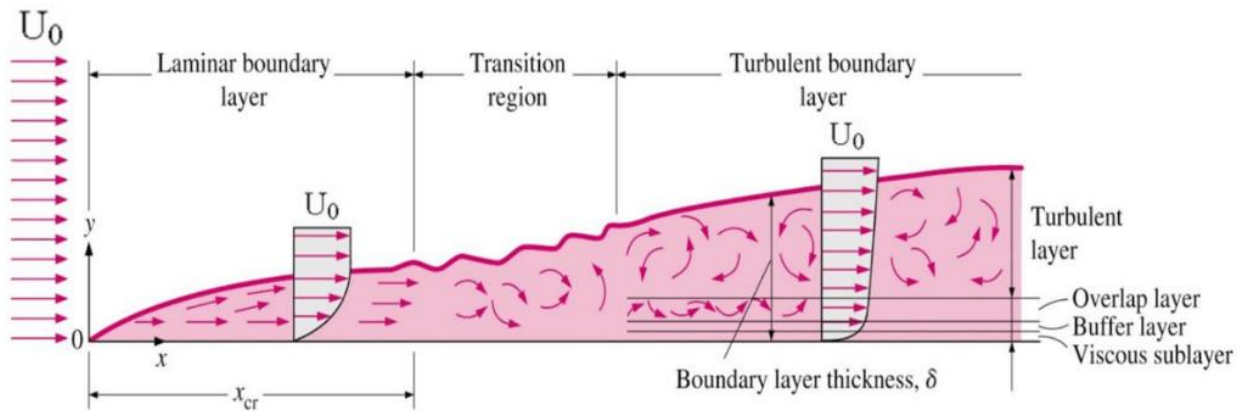
The boundary layer refers to the thin layer of fluid that forms near a solid surface when a fluid flows over it. In this layer, the fluid velocity changes rapidly from zero at the surface to the free stream velocity away from the surface. The boundary layer plays a crucial role in determining the flow characteristics, such as friction drag and heat transfer, in many engineering applications.

The boundary layer thickness is a measure of the extent of this layer from the solid surface. It is defined as the perpendicular distance from the solid surface at which the fluid velocity reaches 0.99 times the free stream velocity (U or U_∞). In other words, it represents the distance from the surface to the point where the flow velocity is approximately 99% of the velocity of the fluid far away from the surface.

The boundary layer thickness varies along the length of the solid surface. It starts from zero at the leading edge and gradually increases as the fluid flows in the direction of the flow. At the trailing edge or the end of the surface, the boundary layer thickness reaches its maximum value. (It is denoted by δ here.)



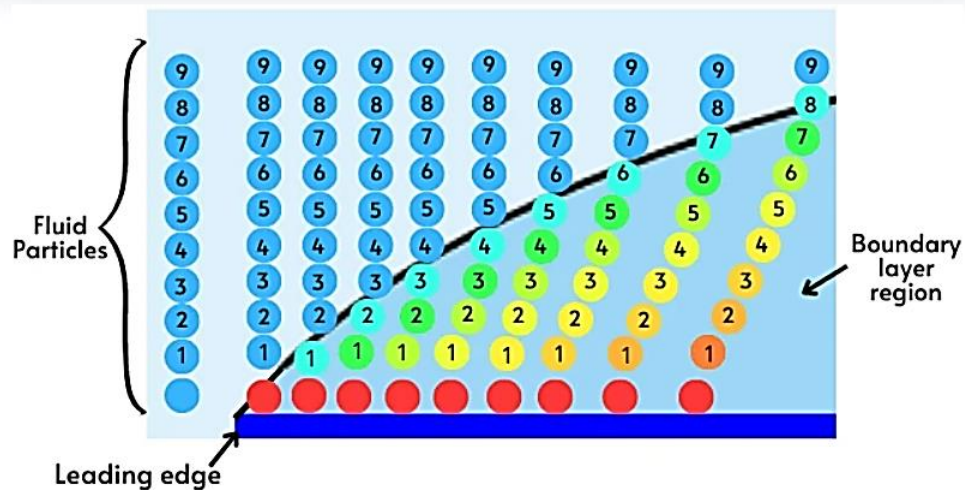
The accompanying diagram illustrates the development of the boundary layer over a flat solid surface. Along the length of the plate, the boundary layer undergoes distinct regions: laminar, turbulent, and a transition zone where the laminar flow transitions into turbulence. Each region is associated with a specific boundary layer thickness. The laminar boundary layer thickness refers to the thickness of the boundary layer in the laminar region, while the turbulent boundary layer thickness represents the thickness in the turbulent region. It's worth noting that within the turbulent zone, there exists a thinner laminar sublayer adjacent to the solid surface, and this sublayer's thickness is denoted as δ' . This laminar sublayer acts as a buffer between the solid surface and the turbulent flow. Overall, understanding the variations in boundary layer thickness and the different regions it encompasses is vital for comprehending and analyzing fluid flow phenomena over solid surfaces.



There are four main characteristics of boundary layer thickness

- At $x = 0$, $\delta = 0$
- At $x = L$, $\delta = \delta_{\max}$
- The boundary layer thickness increases with the increase in distance from the leading edge.
- At $y = \delta$, $\frac{du}{dy} \approx 0$ and $u = 0.99U$

The increase in boundary layer thickness along the length of a solid surface can be attributed to the cumulative effect of fluid viscosity and the interaction between adjacent layers of fluid.



When the fluid particles near the bottom layer come into contact with the leading edge of the flat plate, they adhere to the surface due to the cohesive force between the fluid particles. As the fluid flows over the plate, the bottom layer of fluid particles experiences retardation due to viscosity, which slightly slows down the motion of the particles in that layer.

As the fluid progresses along the surface, this retardation in the bottom layer affects the adjacent upper layer. The particles in the first layer transmit their slower motion to the particles in the second layer, causing them to undergo further retardation. This process continues as the fluid advances, with each layer affecting the one above it.

Consequently, more and more fluid layers enter the region of the boundary layer as they are gradually affected by the viscosity-induced retardation. This accumulation of retardation and transmission of slower motion from one layer to the next results in an increase in the thickness of the boundary layer over the length of the solid surface.

2.2.4.1. Boundary layer thickness for laminar flow

The concept of critical Reynolds number ignores the transition zone. Actually, the critical Reynolds number lies in between the transition zone. For the flow over flat plate, the value of the critical Reynolds number (Re_{cr}) used for the calculation is 5×10^5 .

If $Re_x < Re_{cr}$ then the flow is laminar.

If $Re_x > Re_{cr}$, then the flow is turbulent.

For the laminar boundary layer ($Re_x < Re_{cr}$), the boundary layer thickness by the Blasius solution is given by the below formula,

$$\delta = \frac{4.91x}{\sqrt{Re_x}}$$

2.2.4.2. Boundary layer thickness for turbulent flow

For the turbulent flow ($Re_x > Re_{cr}$), the value of boundary layer thickness based on the Blasius solution is given by the below formula,

$$\delta = \frac{0.37x}{Re_x^{\frac{1}{5}}}$$

Where,

Re_x = Reynolds number at length of x

x = Distance from the leading edge Thickness of Laminar Sublayer

The laminar sublayer is also called a viscous sublayer.

In the region of turbulent flow, the laminar sublayer is the smaller region near the solid surface, where no turbulence existed. In this region, the fluid velocity is only affected by the viscous effects, not by the inertial effects.

The thickness of the laminar sublayer (δ') is given by,

$$\delta' = 11.6 \frac{\text{Kinematic Viscosity}}{\text{Shear Velocity}} = \frac{\nu}{V^*}$$

Where,

$$V^* = \sqrt{\frac{\tau_0}{\rho}}$$

τ_0 =Wall shear stress

2.2.4.3. Factors affecting Boundary layer Thickness

The thickness of the boundary layer is influenced by various factors, and here are a few important ones:

1) Pressure Gradient: The pressure gradient, which indicates the change in pressure along the flow direction, plays a role in determining the boundary layer thickness. If the pressure decreases in the flow direction (negative pressure gradient), it implies an increase in kinetic energy, leading to higher velocities and a thinner boundary layer. Conversely, if the pressure increases in the flow direction (positive pressure gradient), it indicates a decrease in kinetic energy, resulting in deceleration and a thicker boundary layer.

2) Viscosity: Viscosity is a measure of a fluid's resistance to shear or deformation. Higher viscosity causes more resistance to flow, leading to increased retardation of the fluid near the solid surface. As a result, the boundary layer thickness tends to be greater in fluids with higher viscosity compared to those with lower viscosity.

3) Free Stream Velocity: The velocity of the free stream, which refers to the fluid approaching the solid surface, affects the boundary layer thickness. A lower free stream velocity leads to reduced momentum and slower flow, resulting in a thicker boundary layer. Conversely, higher free stream velocities promote faster flow and thinner boundary layers.

4) Distance Along the Surface: The distance traveled along the solid surface also impacts the boundary layer thickness. As the fluid progresses from the leading edge to the trailing edge, the boundary layer thickness generally increases. This is because the layers of fluid closer to the solid surface experience more interactions and retardation, causing a gradual growth in the thickness of the boundary layer.

These factors collectively influence the development and characteristics of the boundary layer, highlighting the complex interplay between pressure gradients, viscosity, fluid velocity, and distance along the solid surface. Understanding these influences is crucial in various engineering applications where boundary layer effects play a significant role.

*** Example for calculating boundary layer thickness**

The air is flowing over a flat plate at the velocity of 3 m/s. If the plate has a length of 3 m, calculate the boundary layer thickness at the trailing edge of the plate.

(Assume, $\nu_{air} = 1.5 \times 10^{-5} = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$)

Given:

$$U = 3 \text{ m/s}$$

$$L = 3 \text{ m}$$

$$\nu_{air} = 1.5 \times 10^{-5} = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$$

Solution:-

At the trailing edge, $x = L$

Thus, the Reynolds number at the trailing edge of the plate is given by,

$$Re_L = \frac{Ux}{\nu_{air}}$$

$$Re_L = \frac{3 * 3}{1.5 * 10^{-5}}$$

$$Re_L = 6 * 10^5$$

As the Reynolds number is greater than the critical Reynolds number ($Re > 5 \times 10^5$), the boundary layer is of turbulent nature.

As per the Blasius solution, the thickness of the turbulent boundary layer is given by,

$$\delta = \frac{0.37x}{Re_x^{\frac{1}{5}}}$$

$$\delta = \frac{0.37 * 3}{(6 * 10^5)^{1/5}}$$

$$\delta = 0.077 \text{ m}$$

2.2.4.4. Pressure Drop Calculations

Calculating pressure drop in heat sink design is essential for optimizing heat transfer efficiency and overall system performance. The pressure drop reflects the resistance to fluid flow through the heat sink, and a higher pressure drop indicates greater flow resistance. By accurately calculating the pressure drop, engineers can make informed decisions regarding heat sink design and geometry to minimize flow resistance while maximizing heat transfer. This ensures efficient thermal management and prevents overheating of electronic components. Moreover, pressure drop calculations aid in selecting appropriate materials and optimizing the size of pumps or fans required to overcome the pressure drop, leading to cost and energy savings. Overall, accurate pressure drop calculations play a crucial role in designing effective heat sinks that provide optimal cooling performance and enhance the reliability and longevity of electronic systems.

There are a number of equations that can be used to calculate the pressure drop in a heat sink. The most common equation is the Darcy-Weisbach equation, which is given by

$$\frac{dp}{dx} = \frac{f}{D} \frac{\rho v^2}{2g}$$

L - length of the heat sink (m)

ρ - density of the fluid (kg/m³)

V - average velocity of the fluid (m/s)

g - acceleration due to gravity (m/s²)

f - Darcy friction factor

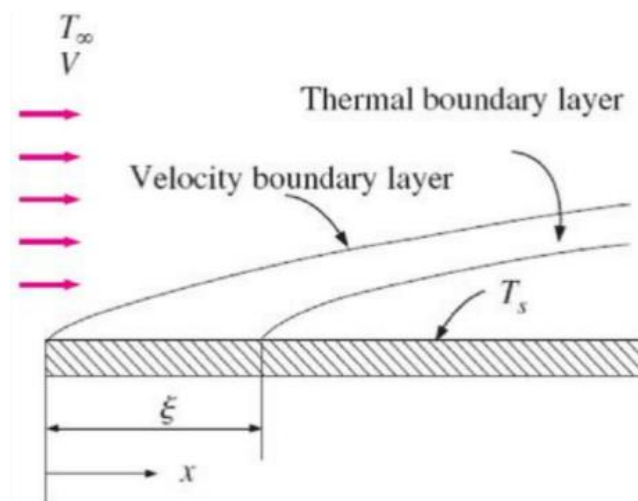
$f = 64/Re_D$ for laminar flow ($Re_D < 2000$)

if flow is turbulent ($Re_D > 2000$) Moody charts or Computational Fluid Dynamics should be used to determine f

* Additional Note :- Nusselt Numbers for various flow scenarios

In heat sink design, it is crucial to determine the appropriate correlation for calculating the average Nusselt numbers in various flow scenarios. These correlations provide relationships between the Nusselt number and relevant parameters such as Reynolds number, Prandtl number, and geometry. By selecting the correct correlation for a specific flow scenario, engineers can accurately predict the convective heat transfer characteristics. These correlations may vary depending on the flow regime (laminar or turbulent), the geometry of the heat sink (such as flat plate or cylindrical fin), and other factors. The proper selection of correlations ensures reliable and precise estimation of average Nusselt numbers, aiding in the design and optimization of heat sinks for effective heat dissipation.

Flow over Flat Plate



$$\overline{Nu}_L = 0.664 Re_L^{1/2} Pr^{1/3} \rightarrow \text{Laminar Flow}$$

$$\overline{Nu}_L = 0.0296 Re_L^{4/5} Pr^{1/3} \rightarrow \text{Turbulent Flow}$$

$$\overline{Nu}_L = (0.037 Re_L^{4/5} - 871) Pr^{1/3} \rightarrow \text{Laminar followed by Turbulent}$$

Flat plate with Unheated length

$$Nu_x = \frac{Nu_x|_{\xi=0}}{[1-(\xi/x)^a]^b} \rightarrow \text{Laminar Flow}$$

$$Nu_x|_{\xi=0} = C Re_x^m Pr^{1/3}$$

	Laminar		Turbulent	
	Isothermal	Isoflux	Isothermal	Isoflux
a	3/4	3/4	9/10	9/10
b	1/3	1/3	1/9	1/9
c	0.332	0.453	0.0296	0.0308
m	1/2	1/2	4/5	4/5

Flow over Cylinder: Churchill and Bernstein Correlation

$$\overline{Nu}_D = 0.3 + \frac{0.62Re_D^{1/2} Pr^{1/3}}{[1 + (0.4/Pr)^{2/3}]^{1/4}} \left[1 + \left(\frac{Re_D}{282,000} \right)^{5/8} \right]^{4/5}$$

Sphere

$$\overline{Nu}_D = 2 + (0.4Re_D^{1/2} + 0.06Re_D^{2/3})Pr^{0.4} (\mu/\mu_s)^{1/4}$$

Isothermal Array of Cylinders

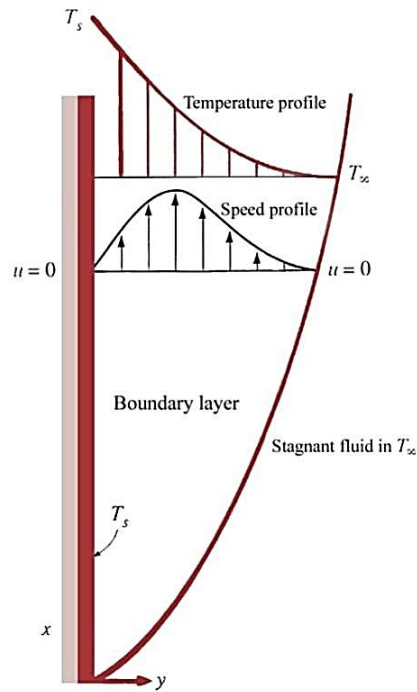
$$\overline{Nu}_D = C_2 [C Re_{D,\max}^m Pr^{0.36} (Pr/Pr_s)^{1/4}]$$

Natural Convection for Vertical Plates

Empirical Heat Transfer Correlations

$$\overline{Nu}_L = 0.68 + \frac{0.670Ra_L^{1/4}}{[1 + (0.492/Pr)^{9/16}]^{4/9}} > \text{Laminar Flow } (Ra_L < 10^9):$$

$$\overline{Nu}_L = \left\{ 0.825 + \frac{0.387Ra_L^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{4/9}} \right\}^2 > \text{Turbulent Flow } 10^9 < Ra_L < 10^{12}$$



Natural Convection for Horizontal flat plate

Facing up

$$\overline{Nu}_L = 0.54 Ra_L^{1/4} \quad (10^4 < Ra_L < 10^7)$$

$$\overline{Nu}_L = 0.15 Ra_L^{1/3} \quad (10^7 < Ra_L < 10^{11})$$

Facing down

$$\overline{Nu}_L = 0.27 Ra_L^{1/4} \quad (10^5 < Ra_L < 10^{10})$$

Horizontal plate

(Surface area A and perimeter p)

(a) Upper surface of a hot plate
(or lower surface of a cold plate)



(b) Lower surface of a hot plate
(or upper surface of a cold plate)



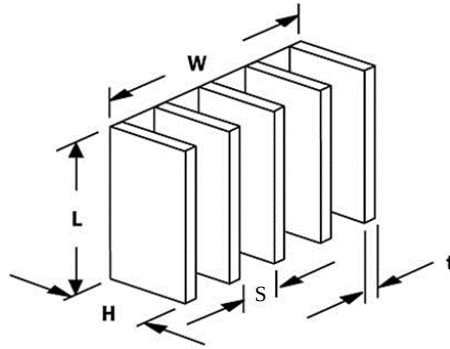
Long Horizontal Cylinder

$$\overline{Nu}_D = \left\{ 0.60 + \frac{0.387 Ra_D^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right\}^2 \quad Ra_D < 10^{12}$$

Sphere

$$\overline{Nu}_D = 2 + \frac{0.589 Ra_D^{1/4}}{[1 + (0.469/Pr)^{9/16}]^{4/9}}$$

Natural Convection through Vertical Channels



Correlation of Bar Cohen and Rohsenow for different boundary conditions

Isothermal plates:

$$\overline{Nu}_S = \left[\frac{C_1}{(Ra_S S/L)^2} + \frac{C_2}{(Ra_S S/L)^{1/2}} \right]^{-1/2}$$

Isoflux plates:

$$Nu_{S,L} = \left[\frac{C_1}{Ra_S^* S/L} + \frac{C_2}{(Ra_S^* S/L)^{2/5}} \right]^{-1/2}$$

$$Nu_{S,L} = \left(\frac{q_s''}{T_{S,L} - T_\infty} \right) \frac{S}{k}$$

$$Ra_S^* = \frac{g \beta q_s'' S^4}{k \alpha \nu}$$

S is the spacing between plates S is maximum when a plate is far apart from another and it's heat transfer is maximum (it is not disturbed by others). If plates are too close, heat plates influence the heat transfer of the adjacent hot plates. But we should find the optimum distance between them to maximize area of heat sink and heat dissipation of it too.

2.2.5. HEAT TRANSFER FROM A VERTICAL WALL BY FREE CONVECTION

When a temperature difference is established between a wall and a stationary fluid, the fluid adjacent to the wall will move upward if the wall temperature is higher than that of the fluid, and downward if the wall temperature is lower. The cause of the movement is the temperature gradient itself. This sets up density gradients in the fluid resulting in buoyancy forces and free convection currents. The rate of heat transfer depends mainly on the fluid motion, and this is very largely governed by a balance between buoyancy and viscous forces; inertia forces play only a small part in free convection.

Consider any fluid element at a temperature θ above the surrounding fluid; the density ρ of this element is smaller than the density ρ_s of the bulk of the fluid, and it is given by $\rho = \rho_s/(1 + \beta\theta)$, where β is the coefficient of cubical expansion. The velocity of this element is a function of the buoyancy force acting on it, which for unit volume is

$$(\rho_s - \rho)g = \beta g \rho \theta$$

Equation 2.2.5.1

It follows that in a solution for the rate of heat transfer by free convection, the group of variables $\beta g \rho \theta$ may be expected to appear. In this section the concepts of the velocity and temperature boundary layers are applied to find an analytical solution for the rate of heat transfer from a vertical wall to a fluid when the flow is laminar.

A velocity and a temperature boundary layer will build up at the wall, as shown in **Fig. 2.2.5.1**. The temperature gradually varies from T_w at the wall to T_s at the edge of the temperature boundary layer where $y = \delta_t$. The velocity gradually varies from zero at the wall to some maximum value in the velocity boundary layer, and then to zero at the edge of this boundary layer where $y = \delta$. It has been found from experiment and confirmed by theory that the thickness of the velocity and temperature boundary layers are about equal (except for very viscous fluids). This is to be expected since the fluid will tend to rise only in the region where the temperature of the fluid is increased.

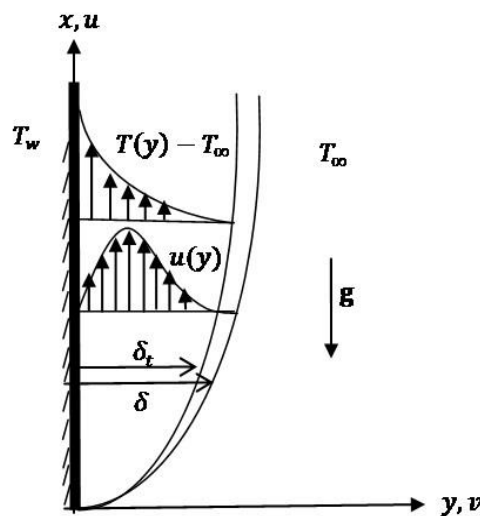


Fig. 2.2.5.1

We can take,

$$\frac{\delta}{x} = 3.93(\text{Pr})^{-1/2} \left(\text{Pr} + \frac{20}{21} \right)^{1/4} (\text{Gr}_x)^{-1/4}$$

Equation 2.2.5.2

Here x is the height above the bottom of the heated wall. Gr_x is a dimensionless group called the Grashof number, and it is equal to,

$$\text{Gr}_x = \frac{\beta g \rho^2 x^3 \theta}{\mu^2}$$

Equation 2.2.5.3

The heat flow at the wall at x is,

$$\dot{q}_w = -k \left(\frac{d\theta}{dy} \right)_0 = -\alpha_x \theta_s$$

Equation 2.2.5.4

where α_x is the local heat transfer coefficient at x . To find α_x it is first necessary to find the temperature gradient $(d\theta/dy)_0$ at the wall. First the temperature profile is expressed in terms of a polynomial in y , and then the temperature gradient is deduced from the polynomial. Let the temperature profile be approximated by a quadratic parabola,

$$T - T_w = \theta = ay + by^2$$

Equation 2.2.5.5

To find the values of the constants a and b , the following boundary conditions may be used:

$$\begin{aligned} \theta &= \theta_s \text{ when } y = \delta \\ \left(\frac{d\theta}{dy} \right) &= 0 \text{ when } y = \delta \end{aligned}$$

Hence (**Equation 2.2.5.5**) becomes,

$$\frac{\theta}{\theta_s} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$$

Equation 2.2.5.6

and therefore

$$\left(\frac{d\theta}{dy} \right)_0 = \frac{2\theta_s}{\delta}$$

Substituting the value of the temperature gradient $(d\theta/dy)_0$ in **(Equation 2.2.5.4)**, we obtain,

$$\alpha_x = \frac{2k}{\delta}$$

Since δ clearly increases with increase of x , this equation implies that α_x decreases with increase of x . If the equation is put into non-dimensional form, we have,

$$Nu_x = \frac{\alpha_x x}{k} = 2 \left(\frac{x}{\delta} \right)$$

Equation 2.2.5.7

The value of δ can be taken from **(Equation 2.2.5.2)**, and hence,

$$Nu_x = 0.509(Pr)^{1/2} \left(Pr + \frac{20}{21} \right)^{-1/4} (Gr_x)^{1/4}$$

Equation 2.2.5.8

In general terms $Nu_x = \phi(Pr, Gr_x)$, suggesting that the Grashof number plays a part in free convection analogous to the part played by the Reynolds number in forced convection.

It is usually of interest to know the average of α over a height l , and this can easily be found by integration. If α is this average value, and α_l is the value at height l ,

then

$$\alpha = \frac{1}{l} \int_0^l \alpha_x \, dx$$

If α_x is found from **(Equation 2.2.5.8)** and substituted in the integral, we find that,

$$\alpha = \frac{4}{3} \alpha_l \text{ or } Nu = \frac{4}{3} (Nu_l)$$

Equation 2.2.5.9

For air, as well as for many other gases, $Pr \approx 0.70$. Substituting this value in **(Equation 2.2.5.8)**, the relation for the local Nusselt number simplifies to

$$Nu_x = 0.376(Gr_x)^{1/4}$$

Equation 2.2.5.10

a horizontal cylinder of diameter d has the same average heat transfer coefficient as a vertical plate of height $l = 2.76d$. Using this result, the expression for the average Nusselt number of the cylinder becomes,

$$Nu_d = 0.527(Pr)^{1/2} \left(Pr + \frac{20}{21} \right)^{-1/4} (Gr_d)^{1/4}$$

Equation 2.2.5.11

where d is the characteristic dimension used in the Nusselt and Grashof numbers. When $Pr = 0.70$, this reduces to,

$$Nu_d = 0.389(Gr_d)^{1/4}$$

Equation 2.2.5.12

In forced convection the Reynolds number is a criterion of whether the flow is laminar or turbulent. The critical Reynolds number above which the flow is turbulent depends on the geometric configuration; thus for flow through tubes the critical value of Re_d is about 2300, and for flow over a plate the critical value of Re_x is about 500000. In free convection the product $(Gr)(Pr)$ serves as a criterion of turbulence, the exact value depending on the geometric configuration. For a vertical plate the flow becomes turbulent when $(Gr_x)(Pr)$ is about 10^9 . The product $(Gr)(Pr)$ itself is a non-dimensional group called the Rayleigh number **Ra**.

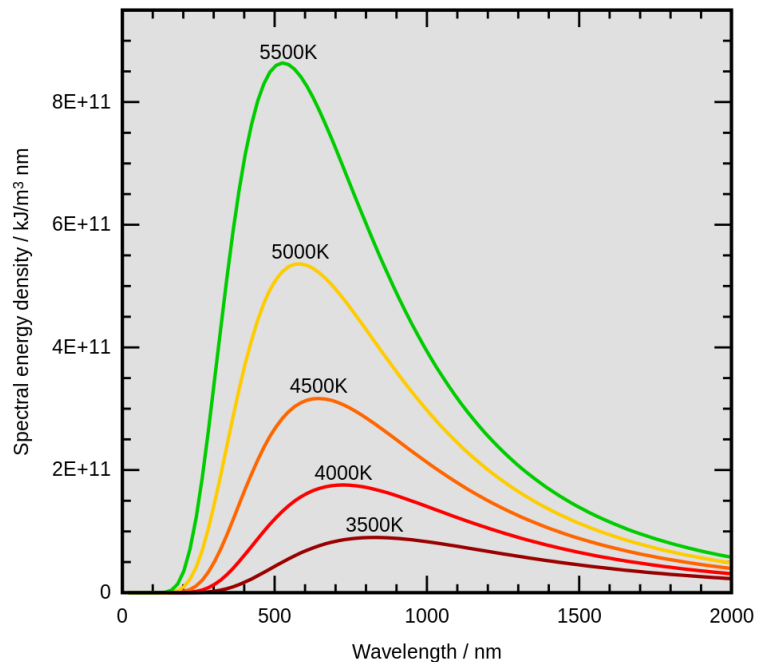
Analytical solutions for free convection problems are usually difficult and even impossible when the flow is turbulent. As in forced convection, the principle of similarity and the method of dimensional analysis are widely used for the systematic correlation of empirical data. In the following section the method of indices is applied to the dimensional analysis of free convection.

2.3. RADIATION PROCESS IN A HEAT SINK

Heat sink is a part applied to a hot electronic device to absorb its heat through conduction and then throw this energy to the ambient environment through convection and radiation.

So here we are focusing on the radiation process associated with a heat sink.

When heat energy is transferred from the hot body to the ambient in the form of electromagnetic waves, it is referred to as the radiation heat transfer.



It's common to ignore how radiation affects a heat sink's design. Numerous sources give a percentage number for the heat lost through a heat sink. The effect of radiation cannot be generalized with a single constant number, like most physics and engineering phenomena.

The impact of radiation on a heat sink's efficiency depends on several variables. An overview of radiation is required before looking into those factors.

- The highest amount of heat (in Watts) that can be radiated from a surface is provided by:

$$Q = \epsilon \sigma A T_s^4 \text{ ----- (Eq. 2.3.1)}$$

Where,

A = surface area of the radiating surface

σ = Stefan-Boltzman constant

T_s = surface temperature in Kelvin (K)

ϵ = emissivity

This surface is considered an ideal radiator or blackbody.

The radiation intensity of a black body refers to the amount of radiation emitted per unit solid angle and per unit area of the black body's surface. The radiation intensity is also known as the spectral radiance.

- According to Planck's law, which describes the spectral radiance of a black body, the radiation intensity at a specific wavelength (λ) and temperature (T) can be calculated using the following equation:

$$I(\lambda, T) = (2\pi hc^2/\lambda^5) * (1/(e^{(hc/\lambda kT)} - 1)) \text{ ----- (Eq. 2.3.2)}$$

Where,

$I(\lambda, T)$ is the radiation intensity (spectral radiance) at wavelength λ and temperature T ,

h is the Planck constant (approximately 6.626×10^{-34} J·s),

c is the speed of light in a vacuum (approximately 3.00×10^8 m/s),

k is the Boltzmann constant (approximately 1.38×10^{-23} J/K),

λ is the wavelength of the radiation,

T is the temperature of the black body.

The above equation gives the radiation intensity as a function of wavelength. It shows that the radiation intensity increases with decreasing wavelength (i.e., higher frequencies) and increasing temperature.

At the same temperature, surfaces that are not perfect radiators emit less energy than a blackbody. A radiative property known as emissivity (ϵ) is applied to these surfaces. The emissivity's value ranges from 0 to 1. When compared to a blackbody, the surface's ability to emit heat is determined by its emissivity.

- Values for common heat sink surface finishes and materials are shown in the following table.

Surface Material	Emissivity
Extruded Aluminum	0.09
Anodized Aluminum	0.85
Unfinished Aluminum Plate	0.09
Copper, commercial burnished	0.07
Copper Polished	0.03
Oil Paints, (various colors)	0.90-0.96

- The equations that depict such interactions are not as simple as the above equation since a heat sink is made up of multiple surfaces that absorb and emit radiation to one another and the enclosure. However, equation's general principle is still valid. It provides a comprehensive explanation of the radiation calculations and associated equations for a plate fin heat sink.

$$Q = \epsilon \sigma A (T_s^4 - T_{sur}^4) \text{ ----- (Eq. 2.3.3)}$$

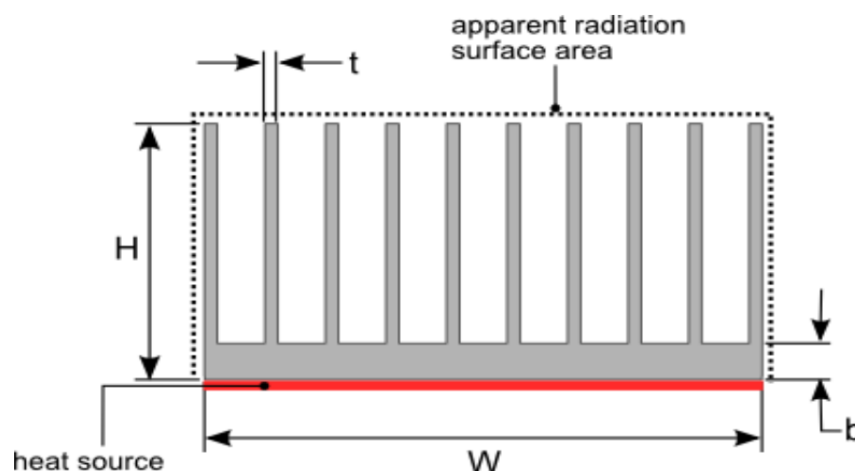
The above computations include a number of equations and can be time-consuming to complete. Calculating the apparent radiation surface area will still allow you to utilize the above equation to obtain an accurate estimate of the heat loss resulting from radiation from your plate fin heat sink. The heat sink is assumed to be a solid block with the same external dimensions in order to calculate the apparent radiation surface area.

The surface area of that block as shown in figure is then calculated using the below equation and used in the above equation. This calculation method does not account for the variation in temperature between the base of the heat sink and tip of the fins which may be significant in forced convection, with long fins, heat sinks made from low conductivity material or a combination of these.

Additionally, the use of the apparent radiation surface area does not accurately account for the surface area of the fins. As such this method should not be used if very accurate results are needed.

$$A_{rad} = 2H(L+W) + LW \text{ ----- (Eq. 2.3.4)}$$

L = length of the heat sink



3. HEAT SINK DESIGN

Heat sinks can be categorized into various design types based on their overall structure and configuration. Here are some common categories of heat sinks:

3.1 DIFFERENTS PIN PROFILE

3.1.1 PIN FINNED

Pin finned heat sinks dissipate heat primarily through convection. The pins increase the surface area, allowing for better heat transfer to the surrounding air. This profile is effective at dissipating heat in low airflow conditions, as it promotes turbulence and enhances convective heat transfer. However, the high pressure drop across the pins can be a disadvantage.

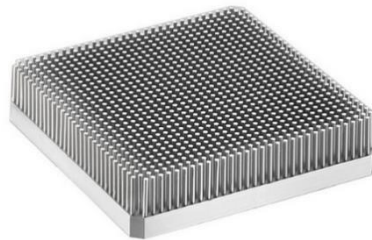


Figure 3.1.1: Pin Finned heat sink

3.1.2 STRAIGHT FINNED

Straight finned heat sinks also rely on convective heat transfer. The straight fins provide increased surface area for heat dissipation, and airflow over the fins carries away the heat. Straight fin profiles are commonly used in applications with moderate airflow. However, they may have limitations in dissipating high heat loads compared to other profiles with larger surface areas.



Figure 3.1.2: Straight Finned heat sink

3.1.3 LOUVERED FINNED

Louvered finned heat sinks offer improved heat dissipation compared to straight fins. The louvered design increases the surface area and promotes better airflow and heat transfer. These fins are effective at dissipating heat in both low and moderate airflow conditions. The louvered profile reduces pressure drop and enhances convective heat transfer, making it a preferred choice for applications requiring efficient cooling in compact spaces.

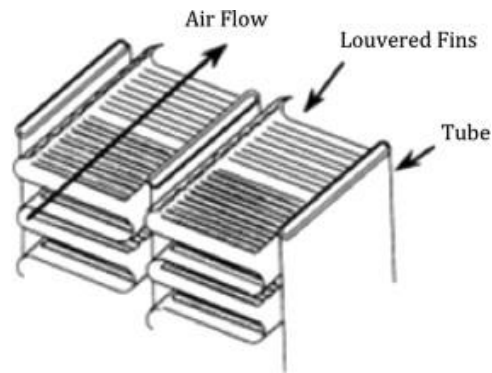


Figure 3.1.3: Louvered Finned heat sink

3.1.4 OFFSET

Offset finned heat sinks dissipate heat through a combination of conduction and convection. The offset arrangement increases the effective surface area and promotes better airflow between the fins. This profile is effective for natural convection cooling, where heat transfer occurs without forced airflow. Offset fins provide enhanced heat dissipation compared to straight fins but may have a higher pressure drop.



Figure 3.1.4: Offset heat sink

3.1.5 WAVY

Wavy finned heat sinks also utilize convective heat transfer. The wavy shape increases the surface area and promotes turbulence in the airflow, enhancing heat dissipation. The wavy profile improves convective heat transfer compared to straight fins, allowing for more efficient cooling. However, manufacturing complexity and increased cost can be drawbacks.



Figure 3.1.5: Wavy heat sink

3.1.6 MICROCHANNEL

Microchannel fin profiles are primarily designed for heat dissipation through forced convection. The numerous small microchannels provide an exceptionally high surface area, promoting efficient convective heat transfer. Microchannel heat sinks are capable of handling high heat fluxes and are often used in applications where compact size and high-performance cooling are crucial. However, manufacturing microchannels can be challenging, and flow blockages within the channels can impact heat dissipation efficiency.

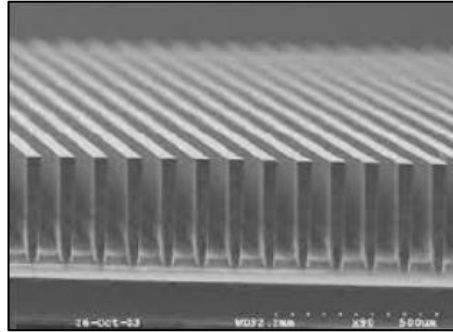


Figure 3.1.6: Microchannel heat sink

3.1.7. COMPARISON BETWEEN DIFFERENT FIN PROFILES

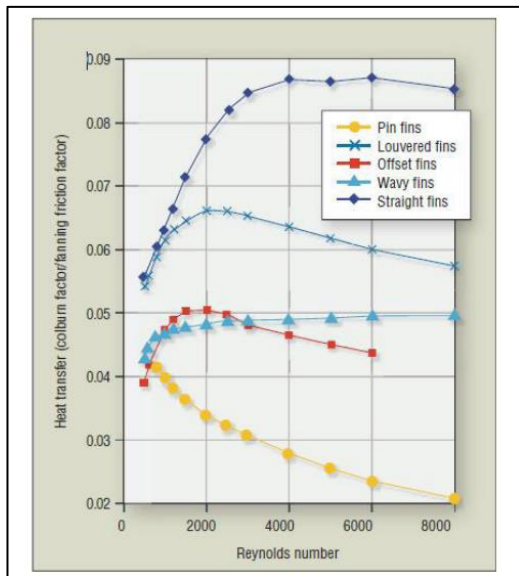


Figure 3.1.7: Profile comparisons based on heat transfer/pressure drop.

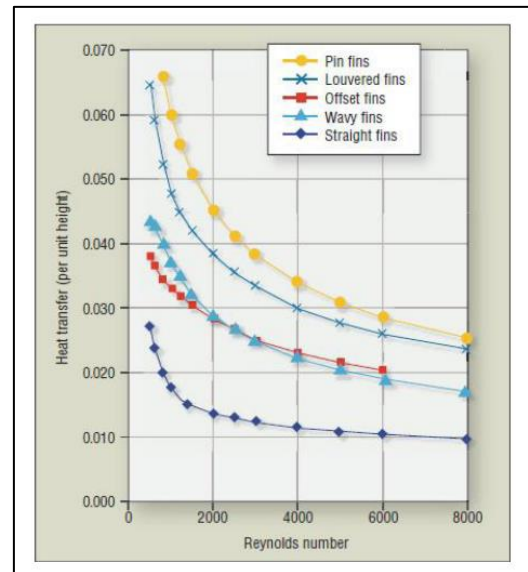


Figure 3.1.8: Profile comparisons based on heat transfer/volume.

3.2 PIN GEOMETRY

Pin geometry in a pin fin heat sink has a significant effect on its heat transfer performance. Here are some key effects of pin geometry:

3.2.1 SURFACE AREA

The surface area of the pins directly impacts the heat transfer capability of the heat sink. A larger surface area allows for more efficient heat dissipation. Different pin geometries offer varying surface area-to-volume ratios, affecting the overall heat transfer performance.

3.2.1 HEAT TRANSFER PATH

The geometry of the pins determines the path that heat follows from the heat source to the surrounding environment. Pin geometries with longer or more convoluted paths can enhance heat transfer by increasing the contact area and promoting turbulence, thereby improving heat dissipation.

3.2.3 AIRFLOW RESISTANCE

The shape and spacing of the pins affect the airflow resistance through the heat sink. Higher pin densities or complex geometries can impede airflow, resulting in increased pressure drop. It is essential to strike a balance between maximizing surface area and maintaining acceptable airflow resistance.

3.2.4 FLOW DISTRIBUTION

Pin geometry influences the distribution of airflow within the heat sink. Different pin shapes and arrangements can affect how air flows between the pins, impacting the cooling efficiency. Optimizing pin geometry can help ensure uniform airflow distribution and minimize hotspots.

3.2.5 TURBULENCE GENERATION

Pin geometries that induce turbulence in the airflow can enhance heat transfer by disrupting the boundary layer. Turbulent flow promotes better mixing and convective heat transfer, improving the overall cooling performance of the heat sink.

3.2.6 MANUFACTURING FEASIBILITY

The complexity and feasibility of manufacturing the chosen pin geometry are important considerations. Some pin geometries may be more challenging to manufacture or may require specialized processes, which can impact cost, production time, and scalability.

3.2.7 WEIGHT AND SPACE CONSIDERATIONS

The chosen pin geometry should align with weight and space constraints within the electronic system. Compact and lightweight pin geometries are desirable in applications where size and weight are critical factors.

3.2.8 STRUCTURAL INTEGRITY

The geometry of the pins influences the mechanical strength and stability of the heat sink. The design should ensure that the pins can withstand vibration, mechanical stress, and thermal expansion/contraction without compromising their structural integrity.

When designing a pin fin heat sink, careful consideration of pin geometry is necessary to optimize heat transfer performance, airflow characteristics, and manufacturability. Computational fluid dynamics (CFD) simulations, prototype testing, and empirical data analysis can help determine the most effective pin geometry for a specific application.

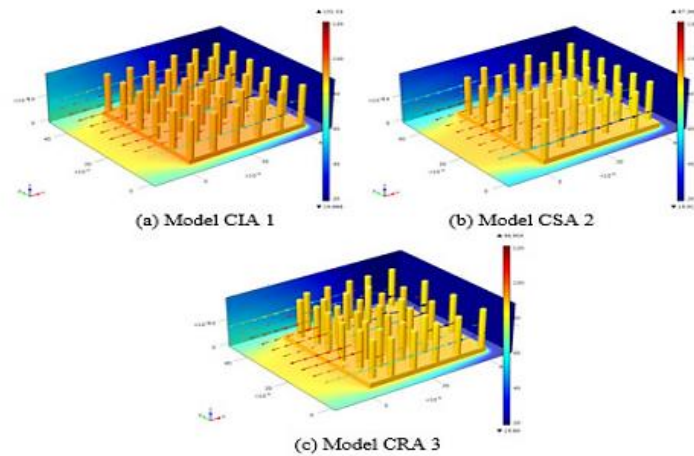


Figure 3.2.1: 3D circular pin fin heat sink models with different arrangement

3.3 PIN DENSITY

Pin density in a pin fin heat sink refers to the number of pins per unit area or volume. It is a crucial parameter that affects the heat transfer performance and overall efficiency of the heat sink. Here is a more analytical description of the effects of pin density:

3.3.1 SURFACE AREA

Pin density directly influences the available surface area for heat transfer. Higher pin densities result in increased surface area, providing more contact points for heat exchange with the surrounding air. This, in turn, enhances the heat dissipation capability of the heat sink.

3.3.2 HEAT TRANSFER AREA

As pin density increases, the total heat transfer area of the heat sink also increases. This leads to improved heat dissipation since a larger surface area allows for more efficient transfer of heat from the heat source to the pins and then to the surrounding air.

3.3.3 AIRFLOW RESISTANCE

Pin density affects the resistance to airflow through the heat sink. Higher pin densities create more obstacles for the airflow, increasing the pressure drop across the heat sink. This can impact the overall system airflow and may require the use of more powerful fans or increased airflow pressure.

3.3.4 THERMAL RESISTANCE

The thermal resistance of the heat sink is inversely proportional to the pin density. Higher pin densities result in lower thermal resistance, allowing for better heat conduction and dissipation. Lower thermal resistance ensures that heat is efficiently transferred away from the heat source.

3.3.5 Pressure Drop

Increasing the pin density also increases the pressure drop across the heat sink. Pressure drop refers to the decrease in air pressure as it flows through the heat sink. Higher pin densities lead to more resistance to airflow, resulting in a higher pressure drop. This needs to be considered when selecting fans or blowers to ensure adequate airflow and cooling performance.

3.3.6 Airflow Distribution

Pin density affects the distribution of airflow within the heat sink. Higher pin densities can cause variations in airflow distribution, resulting in uneven heat dissipation across the heat sink. Proper design and optimization of pin density can ensure uniform airflow distribution, minimizing hotspots and maximizing cooling efficiency.

3.3.7 Manufacturing Complexity

As pin density increases, the manufacturing complexity of the heat sink also increases. Higher pin densities may require more precise machining or manufacturing processes, which can impact cost, production time, and feasibility. It is important to strike a balance between pin density and manufacturing considerations.

Optimizing pin density involves finding the right balance between maximizing surface area and ensuring adequate airflow for efficient heat transfer. Analytical modeling, simulations, and experimental testing can help determine the optimal pin density for a specific application, considering factors such as heat load, available space, airflow requirements, and manufacturing constraints.

3.4 BASE PLATE

The base plate of a heat sink is the component that makes direct contact with the heat source, such as a processor or a power electronic device. Its primary function is to efficiently transfer heat from the heat source to the heat sink's fins, enabling effective heat dissipation. Here are some important aspects to consider when describing the base plate:

3.4.1 MATERIAL SELECTION

The base plate is typically made of materials with high thermal conductivity, such as copper or aluminum. Copper has excellent thermal conductivity, allowing for efficient heat transfer. Aluminum is commonly used due to its good thermal conductivity, lighter weight, and lower cost. The material selection depends on factors like thermal requirements, cost considerations, and weight constraints.

3.4.2 SURFACE FLATNESS

The base plate should have a flat and smooth surface to ensure maximum contact area and minimize thermal resistance. A smooth surface helps to improve thermal conductivity and reduce air gaps between the heat source and the base plate, optimizing heat transfer.

3.4.3 SURFACE FINISH

The surface finish of the base plate can impact its thermal performance. Finishes like nickel plating or anodization can provide a protective layer and improve corrosion resistance. Certain surface treatments, such as copper plating or heat spreading compounds, may also be applied to enhance thermal conductivity.

3.4.4 THERMAL INTERFACE MATERIAL (TIM)

The base plate serves as the interface between the heat source and the heat sink. Thermal interface materials, such as thermal paste, thermal pads, or phase change materials, are often applied between the base plate and the heat source to improve thermal conduction and fill any air gaps. The choice of TIM depends on factors like thermal conductivity, stability, and ease of application.

3.4.5 MOUNTING MECHANISM

The base plate is typically designed to accommodate the mounting mechanism required to secure the heat sink onto the heat source. Common mounting mechanisms include clips, screws, or thermal adhesive films. The design should ensure a secure and reliable attachment, maintaining proper thermal contact.

3.4.6 MANUFACTURING CONSIDERATIONS

The base plate can be manufactured through processes like extrusion, casting, or machining. The manufacturing method should be chosen based on factors like design complexity, cost, and production volume. Machining provides flexibility for intricate designs, while extrusion offers cost-effective production for large quantities.

3.4.7 HEAT SPREADING

The base plate plays a crucial role in spreading the heat evenly across the heat sink's fins. It helps distribute the heat from localized hotspots on the heat source to a larger surface area, facilitating more efficient heat dissipation.

3.4.8 ELECTRICAL ISOLATION (IF REQUIRED)

In some applications, electrical isolation between the heat sink and the heat source is necessary. Insulating materials or coatings can be applied to the base plate to prevent electrical conductivity or shorts.

The base plate design should be optimized to ensure effective thermal contact, low thermal resistance, and efficient heat transfer from the heat source to the rest of the heat sink. Considerations such as material selection, surface flatness, thermal interface materials, mounting mechanism, and manufacturing feasibility are vital to achieve optimal cooling performance.

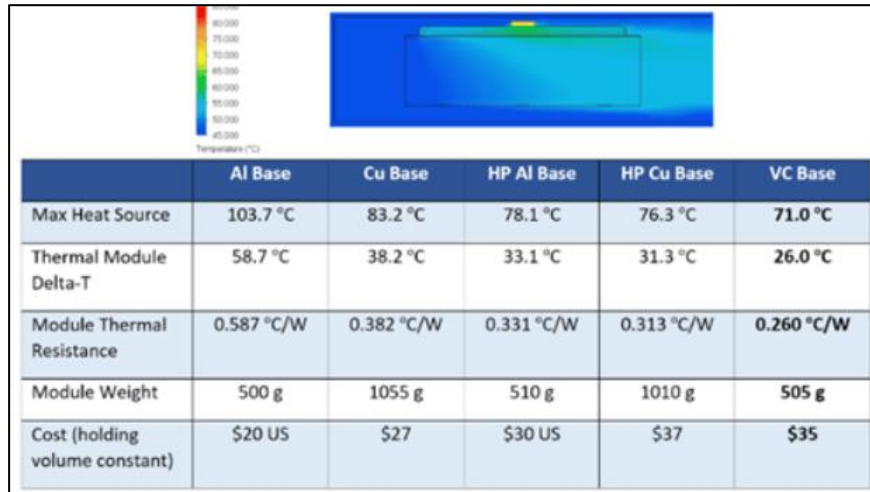


Figure 3.4.1: Base plate configurations

3.5 PIN MATERIAL

The choice of material for the pins in a pin fin heat sink is crucial for its overall performance and heat transfer efficiency. Here are some key considerations and characteristics of pin materials:

3.5.1 THERMAL CONDUCTIVITY

The thermal conductivity of the material directly affects its ability to transfer heat. Higher thermal conductivity allows for more efficient heat transfer from the heat source to the pins and then to the surrounding air. Copper is known for its excellent thermal conductivity, followed by materials like aluminum and various alloys.

3.5.2 COST

Cost considerations play a significant role in material selection. Copper has higher thermal conductivity but is generally more expensive than aluminum. Aluminum is a popular choice for pin fin heat sinks due to its reasonable cost and good thermal conductivity. Other materials like brass or various alloys can also be used, depending on cost constraints and specific application requirements.

3.5.3 WEIGHT

The weight of the material impacts the overall weight of the heat sink. Aluminum is lightweight compared to copper, making it advantageous in applications where weight reduction is a consideration, such as portable electronic devices or aerospace applications.

3.5.4 CORROSION RESISTANCE

The material should be chosen with consideration for the operating environment and the potential for corrosion. Aluminum is susceptible to corrosion, especially in certain environments or when exposed to specific chemicals. Copper is more resistant to corrosion, making it a preferred choice in harsher environments.

3.5.5 MANUFACTURING FEASIBILITY

The chosen material should be compatible with the manufacturing processes used for the heat sink. Aluminum and copper are commonly used materials for pin fins, as they can be easily extruded, machined, or stamped to form the desired pin shapes. Other materials may require more specialized manufacturing techniques.

3.5.6 COMPATIBILITY WITH THERMAL INTERFACE MATERIALS

The material should be compatible with the thermal interface materials (e.g., thermal paste or pads) used between the heat sink and the heat source. It should allow for good thermal contact and minimize thermal resistance at the interface.

3.5.7 ELECTRICAL CONDUCTIVITY

Depending on the application, electrical conductivity may be a consideration. Copper is an excellent conductor of electricity, which can be advantageous in certain electronic systems where electrical grounding or conductivity is important.

When selecting the pin material, it is crucial to consider factors such as thermal conductivity, cost, weight, corrosion resistance, manufacturing feasibility, and application-specific requirements. A comprehensive evaluation of these factors will help determine the most suitable material for the pins in the pin fin heat sink, ensuring efficient heat transfer and optimal cooling performance.

3.6 PIN HEIGHT

Pin height in a pin fin heat sink refers to the distance from the base of the pin to its tip. It is an important parameter that influences the heat transfer characteristics and overall performance of the heat sink. Here are some additional details on pin height:

3.6.1 HEAT DISSIPATION

The height of the pins affects the heat dissipation capacity of the heat sink. Longer pins provide a larger surface area for heat transfer, allowing for more efficient heat dissipation. This is particularly important when dealing with high heat loads or applications that require enhanced cooling performance.

3.6.2 SURFACE AREA

Pin height directly contributes to the total surface area available for heat transfer. Increasing the pin height increases the overall surface area, enabling more contact points for heat exchange between the heat sink and the surrounding air. This leads to improved heat dissipation and lower component temperatures.

3.6.3 AIRFLOW RESISTANCE

Pin height can influence the resistance to airflow through the heat sink. Longer pins may restrict the airflow, resulting in increased pressure drop and reduced overall airflow through the heat sink. It is crucial to balance the pin height with the required airflow to maintain efficient cooling performance.

3.6.4 SPACING AND DENSITY

Pin height is interconnected with pin spacing and density. Increasing the pin height while maintaining the same pin spacing results in a higher pin density, which affects the overall airflow resistance and heat transfer characteristics. Proper optimization of pin height, spacing, and density is necessary to achieve the desired cooling performance.

3.6.5 MANUFACTURING CONSIDERATIONS

The chosen pin height should align with the manufacturing capabilities and constraints. Depending on the manufacturing method employed, there may be limitations on the achievable pin height. Machining processes offer more flexibility in adjusting pin height, while extrusion processes may have specific height limitations based on the die design.

3.6.6 CLEARANCE AND SPACE CONSTRAINTS

The available clearance and space within the electronic system or device can impact the maximum allowable pin height. It is important to consider the available vertical space when determining the optimal pin height to ensure the heat sink fits within the designated area.

3.6.7 STABILITY AND STRUCTURAL INTEGRITY

Longer pins may introduce mechanical challenges related to stability and structural integrity. The design should ensure that the pins are adequately supported and can withstand any mechanical stress or vibrations encountered during operation.

Optimizing the pin height is a critical aspect of pin fin heat sink design. It requires considering factors such as heat load, available space, airflow requirements, and manufacturing feasibility. Analytical modeling, simulations, and prototyping can aid in determining the optimal pin height for a specific application to achieve efficient heat transfer and optimal cooling performance.

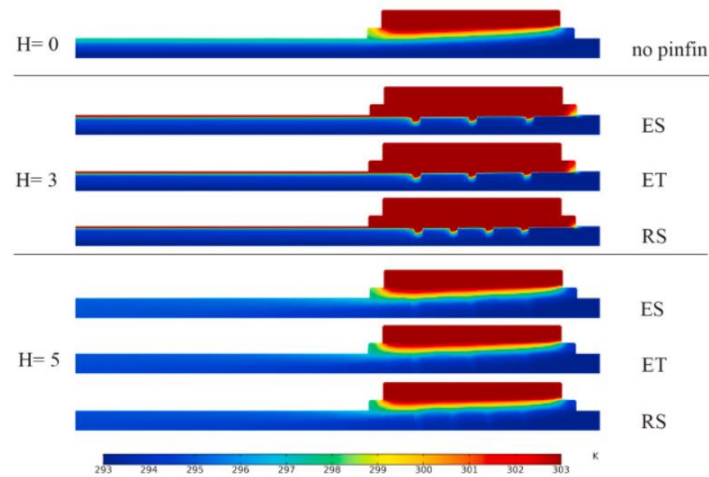
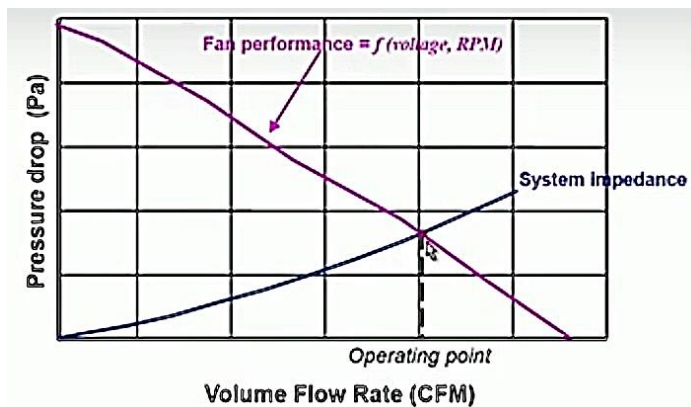


Figure 3.6.1: Variation with pin height

3.7 HEAT SINK CHARACTERIZATION

Heat sink characterization involves evaluating the performance of heat sinks used for cooling electronic components. This process includes wind tunnel tests to measure thermal behavior under controlled airflow. Thermal resistance is a crucial parameter measured by calculating the temperature difference across the heat sink for a given power dissipation. System impedance is assessed to determine the overall resistance offered by the cooling system. Additionally, pressure and volume rate change graphs are used to understand the relationship between the pressure drop and the airflow rate through the heat sink. These graphs provide insights into the heat sink's efficiency and help optimize its design for better heat dissipation.

$$E = \frac{T(s) - T(a)}{Q}$$



$T(s)$: Heat sink base temperature.
 $T(a)$: Approaching air temperature.
 Q : Power supplied to the heat sink

4. MODIFICATION IDEAS FOR HEAT SINK

Heat sinks are widely used in various industries to dissipate heat generated by electronic devices and other systems. While traditional heat sinks have proven effective, there are several modification ideas and new scientific concepts that can enhance their performance. Here are a few:

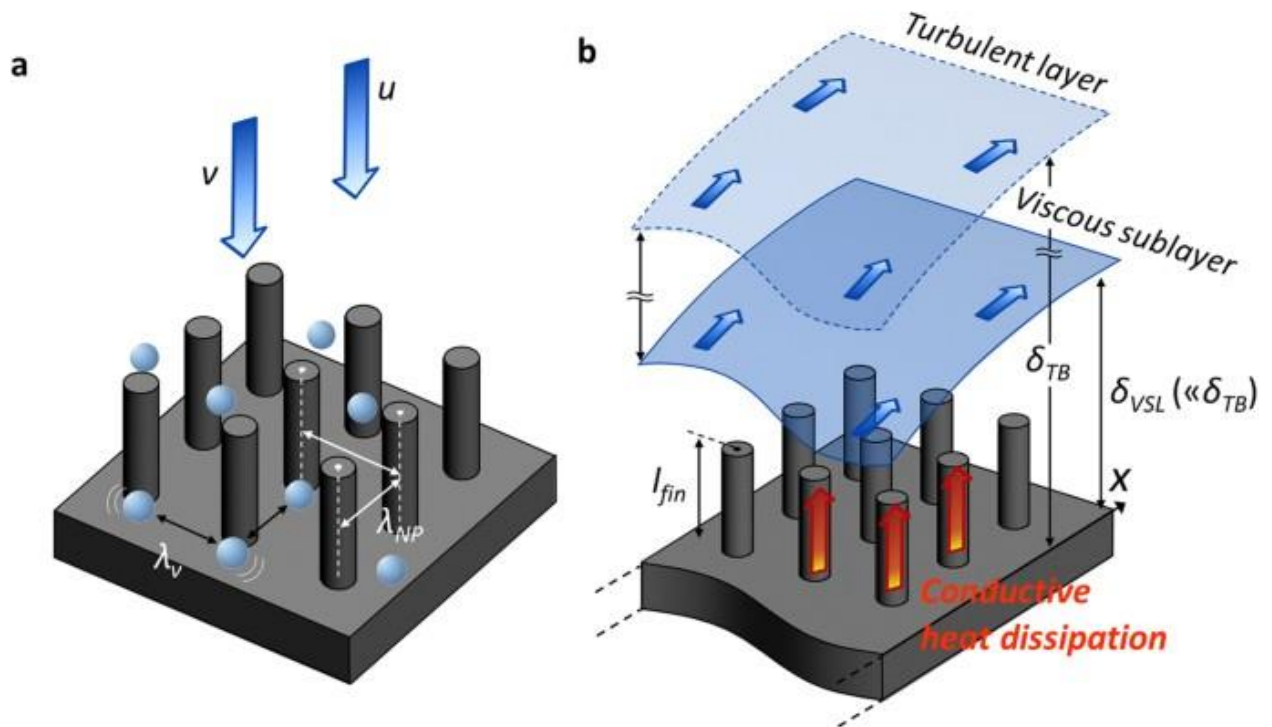
4.1 NANOSTRUCTURED HEAT SINKS

Nanostructured heat sinks involve the integration of nanoscale structures onto the surface of conventional heat sinks. These structures, typically in the form of nanowires, Nano pillars, or Nano fins, provide a significantly increased surface area compared to traditional smooth surfaces. This enlarged surface area allows for improved heat transfer and enhanced cooling efficiency.

The increased surface area-to-volume ratio of nanostructures facilitates efficient heat dissipation by promoting better thermal contact with the surrounding medium, such as air or a coolant fluid. When heat is generated by a device or system, it is transferred to the heat sink through conduction. The nanostructures on the heat sink surface provide more contact points and pathways for heat to move away from the source.

The reduced size of the nanostructures also enhances convective heat transfer. As a fluid, such as air, flows over the heat sink, it experiences increased turbulence due to the presence of nanostructures. This turbulence disrupts the stagnant boundary layer of the fluid and improves heat transfer through convection.

Fabrication techniques like nanolithography or chemical etching are employed to create these nanostructures on the heat sink surface. Nanolithography involves using advanced patterning techniques to precisely shape the desired nanostructures, while chemical etching relies on selectively removing material from the surface to form the desired structures. These processes allow for the controlled design and arrangement of nanostructures, optimizing their effectiveness for heat dissipation.



Nanostructured heat sinks offer several advantages over traditional heat sinks. They provide a larger surface area for heat transfer, enabling efficient cooling in compact systems where space is limited. The enhanced heat dissipation capability reduces the temperature of electronic components, thereby improving their performance, longevity, and reliability. Nanostructured heat sinks are particularly useful in applications with high power density, such as microprocessors, LEDs, power electronics, and other heat-generating electronic devices.

Additionally, advancements in nanomaterials have led to the development of materials with excellent thermal conductivity, such as carbon nanotubes or graphene. These materials can be integrated into nanostructured heat sinks to further enhance their heat dissipation properties. The exceptional thermal conductivity of nanomaterials allows for efficient heat conduction within the heat sink, ensuring rapid and uniform spreading of heat across the surface.

In summary, nanostructured heat sinks leverage nanoscale structures to increase the surface area available for heat transfer, leading to improved cooling performance. These heat sinks find applications in various industries where efficient heat dissipation is crucial, enabling better thermal management and enhancing the overall performance and reliability of electronic systems.

4.2 THERMOELECTRIC HEAT SINKS

Thermoelectric heat sinks are a specialized type of heat sink that not only dissipates heat but also converts a temperature gradient into usable electrical energy through the phenomenon known as the thermoelectric effect. This unique capability makes them suitable for applications where waste heat can be converted into power while simultaneously providing cooling.

The thermoelectric effect is based on the principle that certain materials, known as thermoelectric materials, can generate an electric current when exposed to a temperature gradient. These materials are typically semiconductors with special properties that allow them to exhibit the thermoelectric effect efficiently.

Thermoelectric heat sinks consist of multiple thermoelectric elements, also called thermoelectric modules, integrated into the heat sink structure. Each thermoelectric module comprises two different types of thermoelectric materials: n-type (negative) and p-type (positive) semiconductors. These materials have distinct electrical properties that facilitate the generation of an electric current when there is a temperature difference between the two sides of the module.

The operation of a thermoelectric heat sink involves the following steps:

I. Heat Absorption

The hot side of the thermoelectric module is in direct contact with the heat source, such as a high-power electronic component or a heat-generating system. The thermoelectric module absorbs heat from this source.

II. Temperature Gradient

As the absorbed heat flows through the thermoelectric module, it creates a temperature gradient across the module. The hot side remains in contact with the heat source, maintaining a higher temperature, while the cold side is exposed to a relatively lower temperature.

III. Thermoelectric Effect

The temperature gradient induces a flow of charge carriers (electrons or holes) within the thermoelectric materials. Due to the unique properties of the n-type and p-type semiconductors, the charge carriers move in opposite directions, resulting in an electric current.

IV. Power Generation

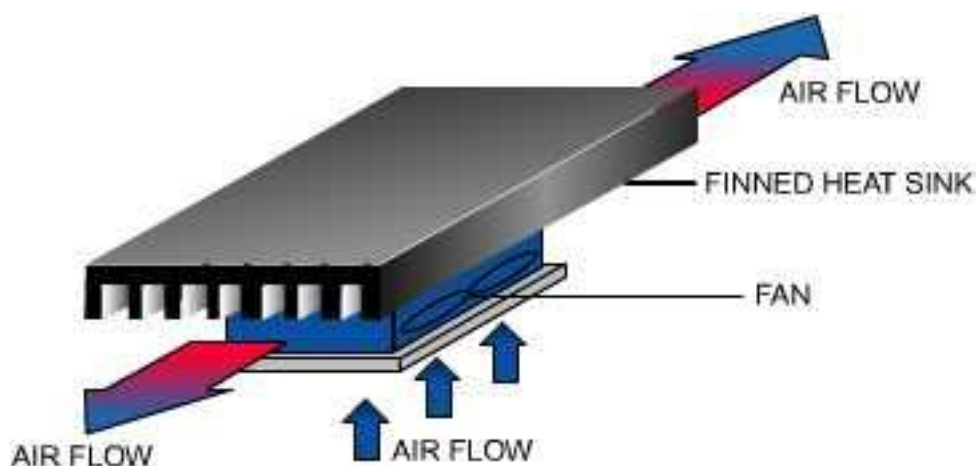
The generated electric current can be harnessed and used to power external devices or stored for future use. This conversion of waste heat into usable electrical energy is a significant advantage of thermoelectric heat sinks.

V. Heat Dissipation

As the electric current is generated, the cold side of the thermoelectric module actively cools down by dissipating heat to the ambient environment or a separate cooling system. This cooling effect helps maintain the desired temperature of the heat-generating component or system.

It's important to note that the efficiency of thermoelectric heat sinks is influenced by various factors, including the properties of the thermoelectric materials, the temperature gradient across the module, and the overall design and optimization of the heat sink system.

Thermoelectric heat sinks find applications in scenarios where both heat dissipation and energy harvesting are required. They are used in diverse fields, such as electronics cooling, automotive industry, aerospace, waste heat recovery, and renewable energy systems. By converting waste heat into usable electricity, thermoelectric heat sinks contribute to improved energy efficiency and sustainability in various applications.



4.3 MICRO FLUID HEAT SINK

Microfluidic heat sinks, also known as microchannel heat sinks or microscale heat sinks, are heat dissipation devices that utilize microfluidic channels to enhance cooling efficiency. These heat sinks are designed to manage and remove heat from high-power electronic components or other heat-generating systems.

Microfluidics involves the manipulation and control of small amounts of fluids within microscale channels typically ranging in size from micrometers to millimeters. Microfluidic heat sinks take advantage of the unique fluidic behavior at this scale to improve heat transfer and cooling performance.

The design of microfluidic heat sinks includes a network of microchannels etched or fabricated on a substrate, usually made of materials with high thermal conductivity, such as copper or silicon. These microchannels are designed to have intricate patterns and high aspect ratios to maximize the surface area available for heat exchange.

➤ The operation of microfluidic heat sinks involves the following steps:

I. Heat Absorption

The microchannels come in direct contact with the heat source, such as a semiconductor device or a high-power electronic component. Heat is transferred from the source to the fluid within the microchannels through conduction.

II. Fluid Flow

A coolant fluid, typically a liquid or gas, is circulated through the microchannels. The fluid absorbs heat from the heat source, increasing its temperature as it flows.

III. Convective Heat Transfer

As the fluid flows through the microchannels, convective heat transfer occurs between the fluid and the channel walls. The increased surface area and the small channel dimensions facilitate efficient heat transfer.

IV. Heat Dissipation

The heated fluid exits the microchannels and is directed towards a heat exchanger, where it releases heat to the ambient environment or is cooled using an external cooling system, such as a radiator or a fan.

V. Fluid Recirculation

The cooled fluid is then reintroduced into the microchannels to repeat the heat dissipation cycle.

- Microfluidic heat sinks offer several advantages for heat dissipation:

I. Enhanced Heat Transfer

The small channel dimensions and increased surface area-to-volume ratio enable efficient convective heat transfer between the fluid and the channel walls. This efficient heat exchange promotes effective cooling of the heat-generating component.

II. Uniform Cooling

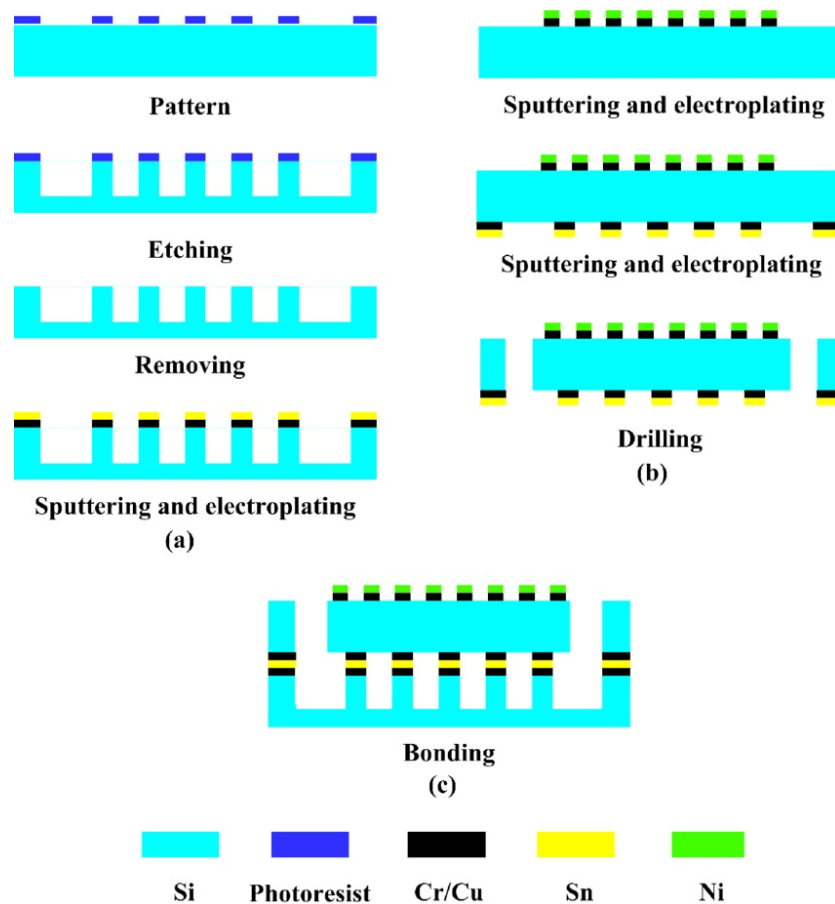
Microfluidic heat sinks provide uniform temperature distribution across the heat-generating surface due to the even flow of coolant through the microchannels. This prevents localized hotspots and helps maintain the desired operating temperature of the component.

III. Compact Size

Microfluidic heat sinks are compact and can be designed to fit into small spaces, making them suitable for applications where size and weight constraints are critical.

IV. Flow Control

The flow rate and direction of the coolant fluid can be precisely controlled, allowing for fine-tuning of the cooling performance. This control enables customized thermal management for specific applications.



Microfluidic heat sinks find applications in various industries, including electronics cooling, power electronics, laser diodes, microprocessors, and microelectromechanical systems (MEMS). They are particularly beneficial in high-power and high-heat-flux applications where traditional cooling methods may be insufficient.

Overall, microfluidic heat sinks provide an efficient and compact cooling solution, leveraging the benefits of microfluidics to effectively dissipate heat from heat-generating components, thereby improving their performance, reliability, and longevity.

4.4 PHOTONIC HEAT SINK

Photonic heat sinks are a novel approach to heat dissipation that utilize the principles of photonics to enhance cooling efficiency. Unlike traditional heat sinks that rely primarily on thermal conduction and convection, photonic heat sinks exploit the control and manipulation of light to optimize heat transfer and radiation.

The concept behind photonic heat sinks involves the strategic design and engineering of materials and structures to control the radiative properties of the system. Radiative heat transfer occurs through the emission and absorption of electromagnetic radiation, particularly in the form of infrared (IR) radiation. Photonic heat sinks aim to maximize the emission of heat as IR radiation while minimizing the absorption of heat from the environment.

➤ The design of photonic heat sinks typically incorporates the following features:

I. Emissive Materials

Photonic heat sinks employ materials that have high emissivity in the IR spectrum. These materials efficiently radiate heat as IR photons, allowing for effective heat dissipation. Selective coatings or surface treatments can be applied to enhance the emissivity of the heat sink.

II. Photon Manipulation

Photonic structures, such as nanostructures or microstructures, can be integrated into the heat sink to manipulate the behavior of photons. These structures control the emission and absorption of IR radiation by modifying the optical properties, such as reflectivity and transmissivity, of the heat sink's surface.

III. Photonic Bandgap Engineering

By designing the photonic structures at specific length scales, it is possible to create a photonic bandgap within the heat sink material. This bandgap restricts the propagation of certain wavelengths of IR radiation, effectively reflecting and trapping heat within the heat sink. This mechanism prevents the reabsorption of radiated heat by the surrounding environment.

IV. Optical Filters

Optical filters can be incorporated into the heat sink to selectively transmit or reflect specific wavelengths of light. These filters can be designed to allow the transmission of visible light, which does not contribute significantly to heat transfer, while reflecting or absorbing IR radiation, which carries the majority of thermal energy.

➤ The benefits of photonic heat sinks include:

I. Improved Thermal Management

By manipulating the behavior of photons, photonic heat sinks can enhance the radiative heat transfer process. This results in improved cooling efficiency, especially in situations where traditional conduction and convection methods may be limited.

II. Reduced Thermal Resistance

The ability of photonic heat sinks to optimize radiative heat transfer can help reduce the overall thermal resistance of the system. This allows for more effective heat dissipation and helps maintain lower temperatures for heat-generating components.

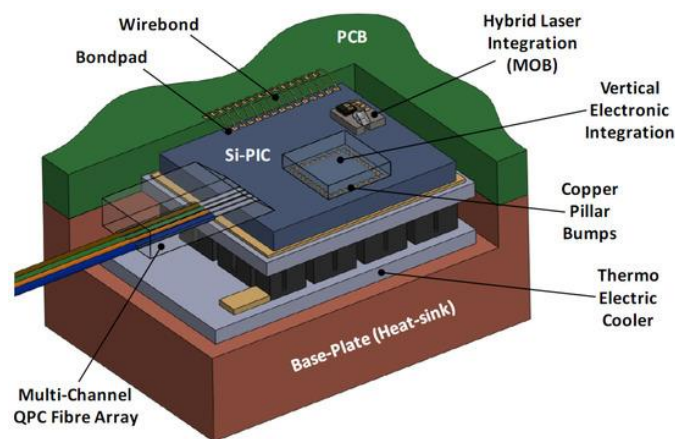
III. Compact Design

Photonic heat sinks can be designed to be compact and lightweight, making them suitable for applications where space and weight constraints are critical.

IV. Customizability

The properties of photonic heat sinks can be tailored and optimized for specific applications. The choice of materials, structures, and coatings can be adjusted to meet the requirements of different heat sources and environmental conditions.

Photonic heat sinks have promising applications in various fields, including electronics cooling, photonics, optoelectronics, and high-power laser systems. They offer a new approach to thermal management, leveraging the principles of photonics to achieve enhanced heat dissipation and efficient cooling of heat-generating components.



4.5 GRAPHENE-BASED HEAT SINKS

Graphene-based heat sinks utilize the extraordinary properties of graphene, a single layer of carbon atoms arranged in a two-dimensional honeycomb lattice, to enhance heat dissipation and improve thermal management. Graphene's exceptional thermal conductivity and unique structural characteristics make it a promising material for heat sink applications.

- The key features and advantages of graphene in heat sinks are as follows:

I. High Thermal Conductivity

Graphene exhibits an exceptionally high thermal conductivity, far surpassing traditional materials like copper and aluminum. It can efficiently conduct heat across its two-dimensional lattice, enabling rapid spreading and dissipation of heat.

II. Thinness and Low Thermal Resistance

Graphene is an atomically thin material, allowing for ultra-thin heat sink designs. Its thinness minimizes thermal resistance and ensures efficient heat transfer from the heat source to the surrounding environment or cooling system.

III. Large Surface Area

Graphene's two-dimensional structure provides an extensive surface area for heat transfer. This large surface area allows for effective contact with the heat source and enhances convective heat dissipation when coupled with a fluid or airflow.

IV. Lightweight and Compact Design

Graphene's exceptional mechanical strength combined with its low weight makes it an ideal material for lightweight and compact heat sink designs. This is particularly advantageous in applications where size and weight constraints are critical.

- There are several approaches to incorporating graphene into heat sink designs:

I. Graphene Coatings

Graphene can be coated onto the surface of conventional heat sink materials, such as copper or aluminum, to enhance their thermal properties. The graphene coating improves thermal conductivity and provides a more efficient path for heat transfer.

II. Graphene Composite Materials

Graphene can be integrated into composite materials, such as polymer composites or metal matrix composites, to enhance their thermal conductivity. These graphene-based composites can be used to fabricate heat sink structures with improved heat dissipation capabilities.

III. Graphene Nanoribbons or Nanoflakes

Graphene nanoribbons or nanoflakes can be patterned and integrated into the heat sink structure to increase the surface area available for heat transfer. These nanostructures provide additional thermal pathways and enhance heat dissipation efficiency.

IV. Graphene Aerogels

Graphene aerogels, three-dimensional porous structures composed of interconnected graphene sheets, can be utilized as heat sink materials. Their high porosity and low density make them excellent candidates for efficient heat dissipation while maintaining lightweight designs.

➤ Graphene-based heat sinks offer several benefits:

I. Improved Heat Transfer

The high thermal conductivity of graphene enables efficient heat transfer from the heat source to the heat sink, minimizing temperature gradients and hotspots. This improves the overall cooling efficiency and thermal performance of the system.

II. Enhanced Thermal Stability

Graphene's excellent thermal stability allows it to withstand high temperatures without degrading or losing its thermal conductivity. This property makes it suitable for demanding heat sink applications.

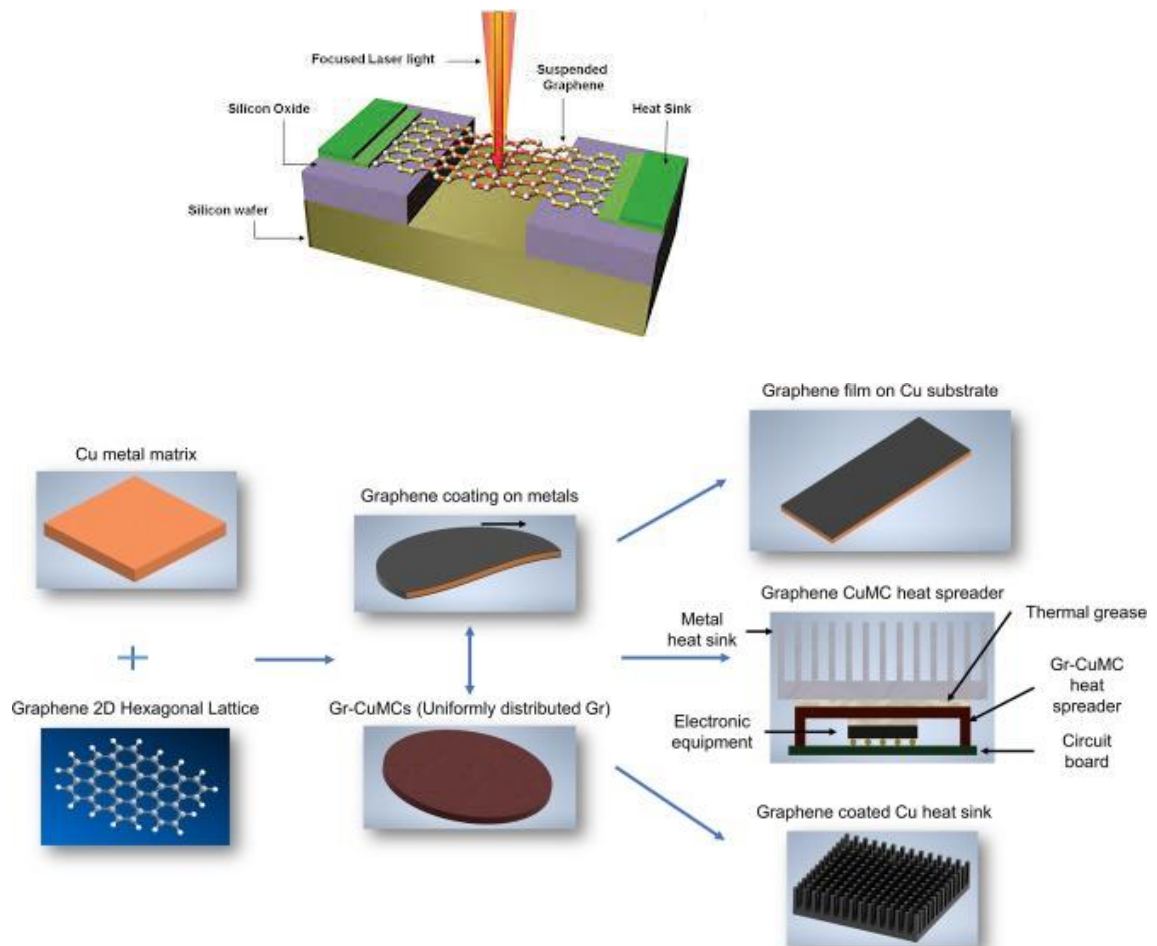
III. Compatibility with Electronics

Graphene is compatible with electronics manufacturing processes, making it feasible to integrate graphene-based heat sinks into electronic devices and systems.

IV. Scalability and Production Potential

Graphene production techniques have been advancing, allowing for scalability and cost-effective manufacturing. This paves the way for large-scale implementation of graphene-based heat sinks in various industries.

Graphene-based heat sinks have applications in areas such as electronics cooling, power electronics, aerospace, and high-performance computing, where efficient thermal management is critical. By leveraging graphene's exceptional thermal conductivity and unique properties, these heat sinks offer the potential for improved heat dissipation, reduced operating temperatures, and enhanced overall system performance.



4.6 MAGNETIC HEAT SINKS

Magnetic heat sinks are innovative cooling systems that utilize the magnetocaloric effect to dissipate heat. The magnetocaloric effect is a phenomenon where certain materials experience temperature changes when subjected to magnetic field variations. Magnetic heat sinks leverage this effect to actively remove heat from a heat source.

➤ The operation of magnetic heat sinks involves the following steps:

I. Magnetocaloric Materials

Magnetic heat sinks are constructed using materials known as magnetocaloric materials. These materials undergo temperature changes in response to alterations in the applied magnetic field. Common magnetocaloric materials include magnetic alloys such as gadolinium, manganese-based compounds, and rare-earth materials.

II. Heat Absorption

The heat sink is placed in contact with the heat source, and heat is transferred from the source to the magnetocaloric material through conduction.

III. Magnetic Field Variation

An external magnetic field is applied or varied around the heat sink, causing the magnetocaloric material to undergo temperature changes. When the magnetic field is increased, the material absorbs heat from the heat source, resulting in a decrease in temperature.

IV. Heat Dissipation

As the magnetocaloric material absorbs heat, it cools down. This cooled material is then brought into contact with a cooler medium, such as air or a liquid coolant, allowing the heat to be transferred and dissipated from the heat sink.

V. Magnetic Field Removal

After the heat is transferred to the cooler medium, the external magnetic field can be removed or reduced. This causes the magnetocaloric material to return to its initial state, ready to absorb heat again.

➤ The benefits of magnetic heat sinks include:

1. Active Cooling

Magnetic heat sinks provide active cooling, where heat is actively transferred away from the heat source using the magnetocaloric effect. This results in efficient heat dissipation and temperature reduction.

2. Compact Design

Magnetic heat sinks can be designed to be compact and lightweight, making them suitable for applications where space and weight constraints are critical.

3. Noiseless Operation

Magnetic heat sinks operate without the need for mechanical components like fans or pumps, resulting in noiseless operation.

4. Scalability

Magnetic heat sinks can be scaled up or down depending on the size and heat dissipation requirements of the application.



Magnetic heat sinks have potential applications in various fields, including electronics cooling, energy-efficient refrigeration, and thermal management of high-power devices. They offer an alternative cooling solution that can provide efficient heat transfer and contribute to improved energy efficiency in heat dissipation systems. However, it is worth noting that the practical implementation of magnetic heat sinks requires careful design considerations and appropriate selection of magnetocaloric materials to optimize their cooling performance.

4.7 Phase Change Materials (PCMs)

Incorporating phase change materials into heat sinks can enhance their thermal performance. PCMs absorb heat during phase transition (solid to liquid or liquid to gas) and release it when they revert to their original state. By integrating PCMs within the heat sink, the absorbed heat can be stored and dissipated over a longer period, improving the overall cooling efficiency.

4.8 Hybrid Heat Sinks

Combining multiple cooling technologies can lead to improved heat dissipation. For example, integrating a conventional heat sink with thermoelectric elements or microfluidic channels can offer enhanced cooling capabilities. Hybrid heat sinks leverage the advantages of each technique to achieve better overall thermal performance.

4.9 Shape Memory Alloy Heat Sinks

Shape memory alloys (SMAs) exhibit unique properties that can be leveraged for heat sinks. SMAs can undergo reversible shape changes in response to temperature variations. By integrating SMAs into heat sinks, it is possible to design adaptive cooling systems that can actively adjust their shape and surface area to optimize heat dissipation.

5.CONCLUSION

In conclusion, our thermodynamic analysis of the heat sink has provided valuable insights into the mechanisms of heat transfer, including conduction, convection, and radiation. By thoroughly examining the design and modification ideas for the heat sink, we have gained a deeper understanding of the principles underlying its functionality and efficiency.

Through our analysis, we have confirmed that the heat sink plays a crucial role in dissipating heat generated by various electronic devices, ensuring their optimal performance and longevity. We observed that heat transfer by conduction occurs primarily within the heat sink material, facilitated by its high thermal conductivity. This mechanism allows for the efficient transfer of heat from the heat source to the heat sink.

Furthermore, our investigation revealed the significance of convection in enhancing heat dissipation. The implementation of fin structures within the heat sink design significantly increases the surface area available for convective heat transfer. This enables better air circulation and improved cooling performance, effectively reducing the overall temperature of the system.

Additionally, we explored the impact of radiation in heat transfer. While radiation is generally less significant compared to conduction and convection in heat sink applications, it becomes increasingly important at higher temperatures. Understanding the radiative properties of the materials used in the heat sink construction is crucial to optimize its thermal performance.

During our analysis, we proposed several modification ideas to enhance the heat sink's effectiveness. These ideas included optimizing fin geometry, improving thermal interface materials, and implementing advanced cooling techniques such as heat pipes or liquid cooling systems. These modifications have the potential to further enhance heat dissipation and reduce the overall operating temperatures of electronic devices.

In conclusion, our thermodynamic analysis has deepened our understanding of heat sink operation and its associated heat transfer mechanisms. By considering the principles of conduction, convection, and radiation, we have gained insights into the various factors influencing the heat sink's performance. Our exploration of design modifications and scientific ideas has provided valuable avenues for future research and development in the field of thermal management.

Overall, this analysis has highlighted the importance of efficient heat sink design and optimization for electronic devices, contributing to their reliability, performance, and longevity. Further research in this area holds great potential for advancements in thermal management, enabling the development of more robust and efficient cooling solutions for a wide range of applications.

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